August 2015

When agents lose their principals

Fixed income liquidity revisited

Matt King
Credit Products Strategist
matt.king@citi.com
+44 207 986 3228

Agenda

Assessing fixed-income liquidity

Regulation and structural change

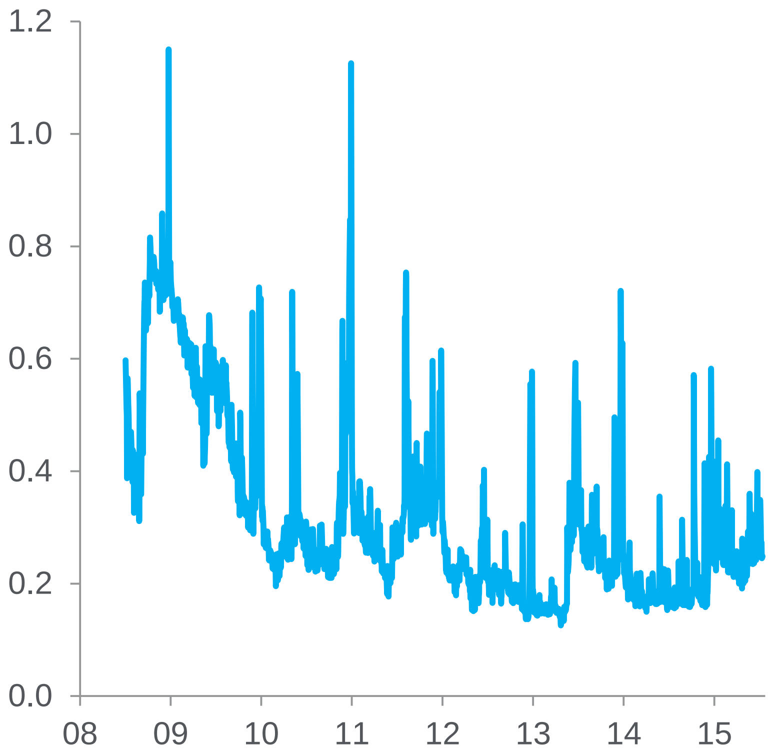
Liquidity in financial shocks

Best practices and innovation

On some metrics, liquidity has seldom been better

UST trading costs lower, if spiky

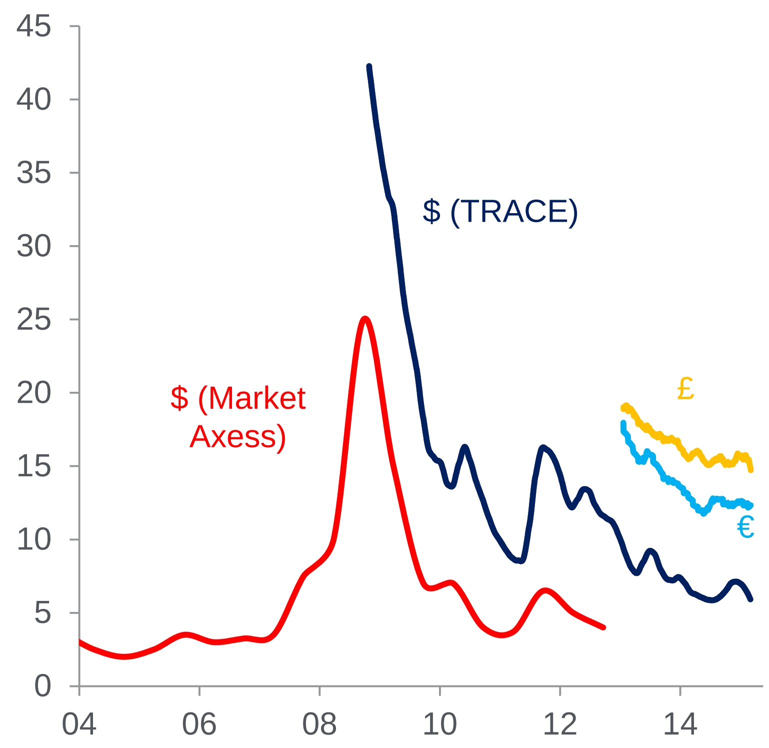
Cost to trade 5k UST contracts, bp of yield



Source: Bloomberg.

Credit bid-offer at post-crisis lows

bp of spread



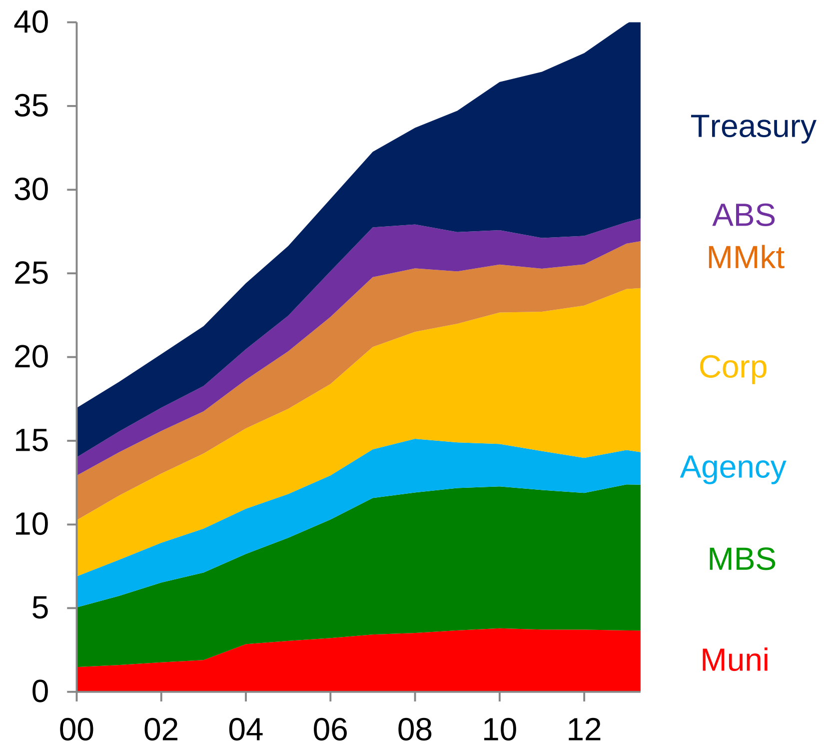
Source: Market Axess BASI, Citi Research.

Much of the time, the cost of trading is low

But markets are turning over less often

A doubling in UST outstandings…

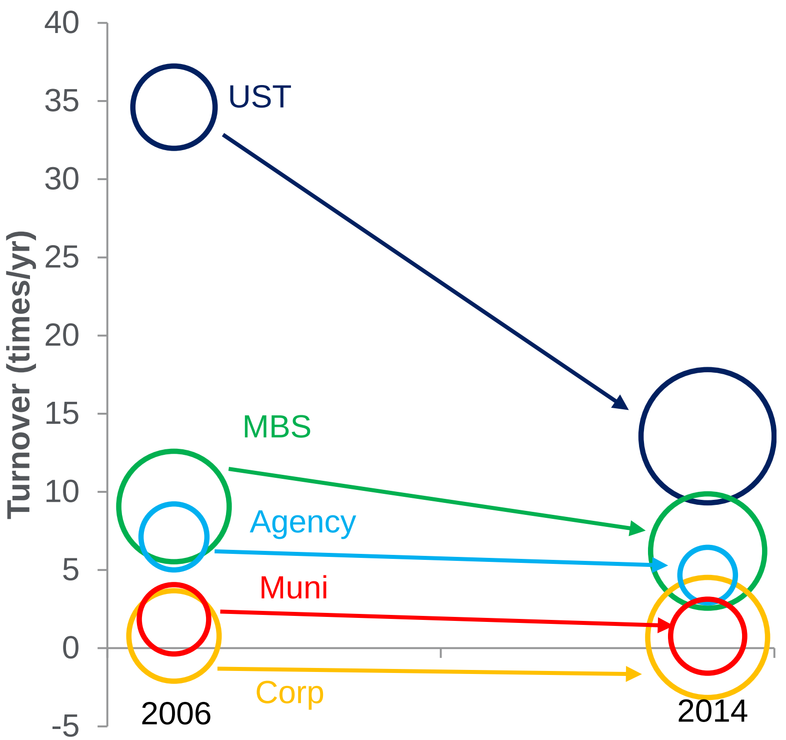
US fixed income outstandings, $tn



Source: SIFMA.

…only partly explains the drop in turnover

Turnover* and outstandings (size), 2006-2014, ex Fed holdings



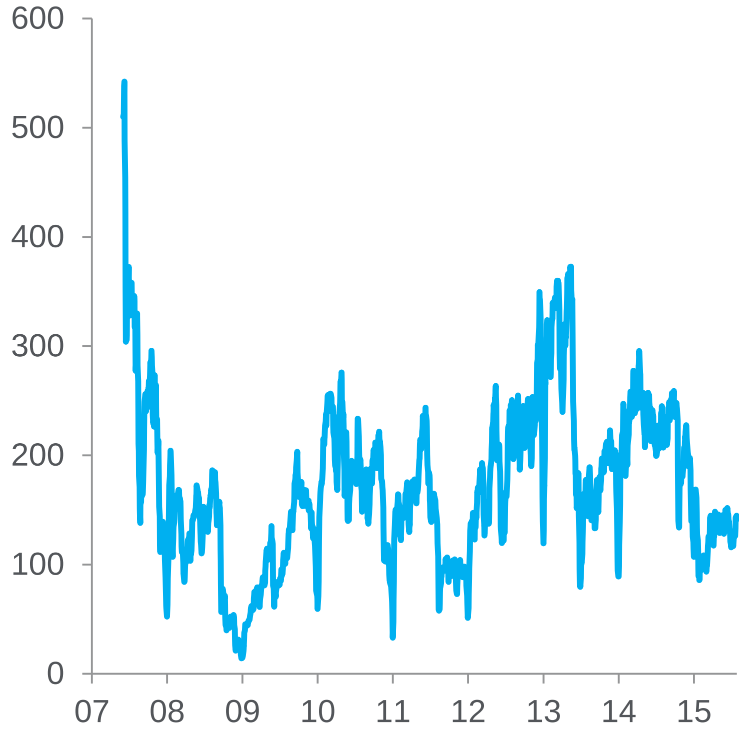
Source: SIFMA, NY Fed primary dealer statistics.
*Omits HFT and other trades not through primary dealers: real decline may be smaller, esp for USTs. However, turnover on Bunds/gilts/JGBs also shows a sharp decline.

The more liquid the market, the more trading has suffered

Depth has fallen

Reduced depth even in on-the-run USTs

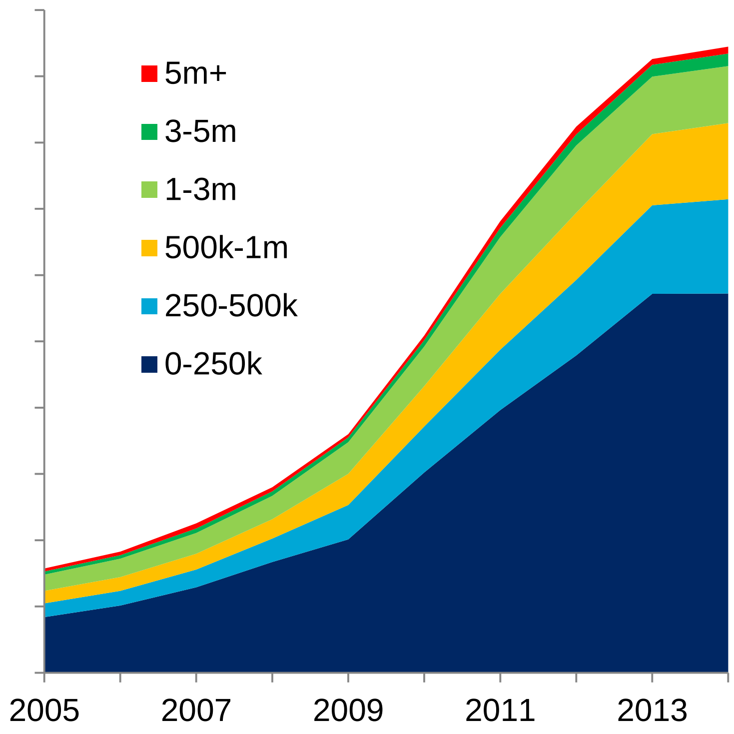
5-day average size of tightest three bids and offers each day, on-the-run 10y UST, $m



Source: JPMorgan, Bloomberg.

E-trading is mostly in odd-lots

Market Axess US credit inquiry for a large US dealer by size, scale removed



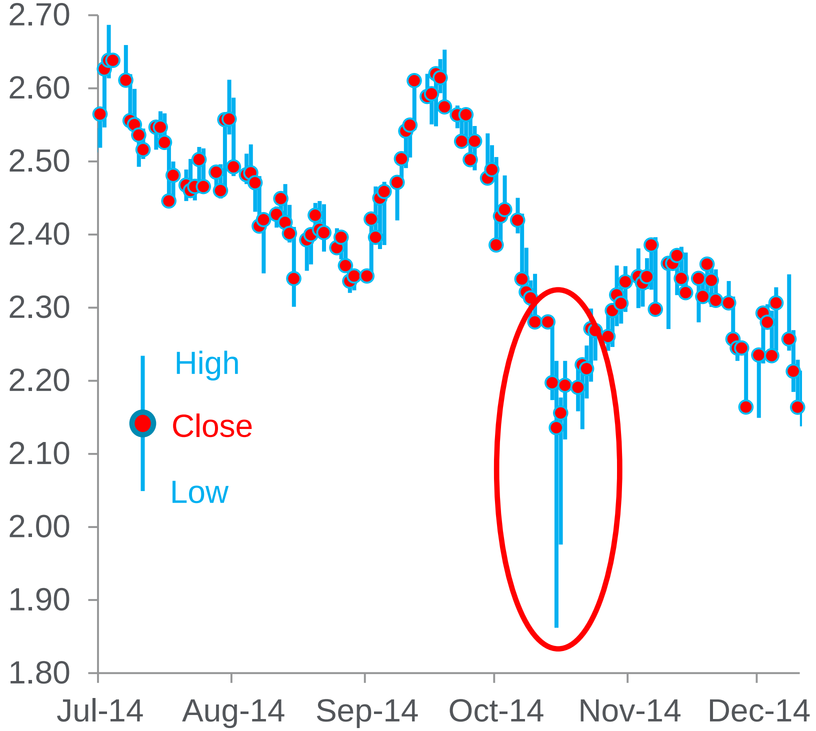
Source: Citi Research.

It is harder to execute large trades

Above all, an increased tendency for liquidity to disappear

A 37bp intraday move in Treasuries

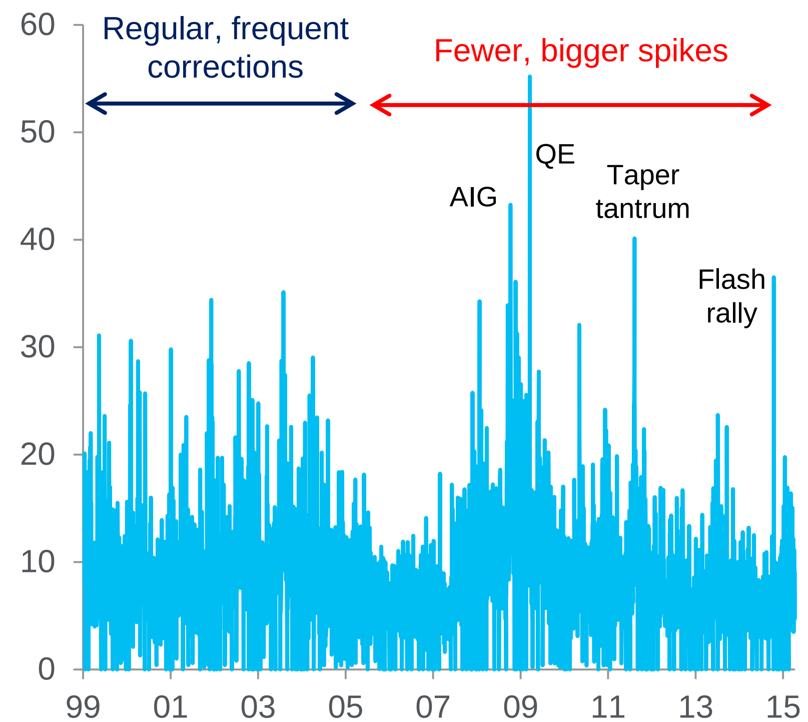
10y UST yield and intraday range, %



Source: Bloomberg.

Previous spikes had obvious causes

Intraday trading range in 10y US Treasuries, bp



Source: Bloomberg.

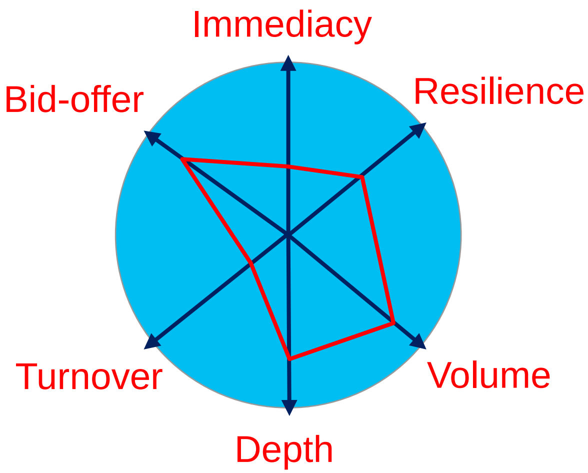
Unexpected bouts of volatility with little obvious explanation

What do we mean by liquidity?

Easily defined

“The ability to transact quickly without exerting a material effect on prices.”

But think less in terms of metrics



And more in terms of preconditions

“Liquidity is optimally achieved when myriad buyers and sellers are ready and willing to trade.”

“…enhanced by market-makers and speculators alike…”

“Liquidity exists when investors are confident in their ability to transact and where risks are quantifiable.”

Source: “Market Liquidity: Definitions and Implications”, Fed Governor Kevin Warsh, March 2007.
See also Borio, C., *Market liquidity and stress: selected issues and policy implications*, BIS (2000).

Liquidity is created when heterogeneous participants can trade together with confidence

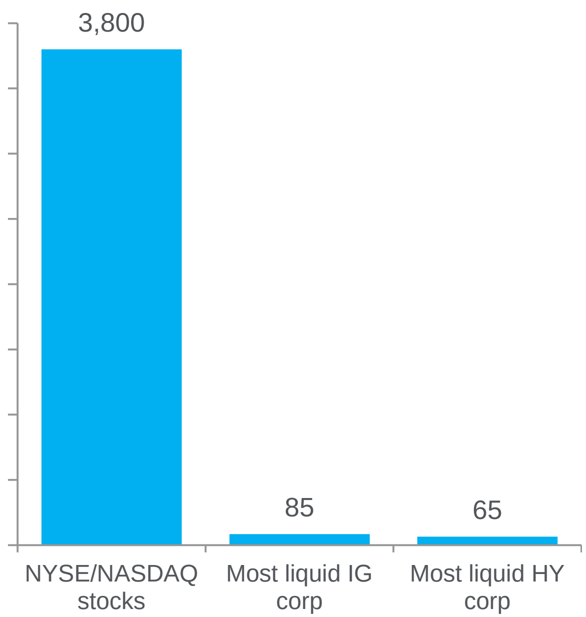
Why fixed income and equities are different

Many more bonds than stocks
Securities in major equity indices

			Unique securities	Mkt cap ($tn)
US	S&P 500	Equities	500	19.4
		Bonds	11561	3.3
	Russell 1000	Equities	1000	22.5
		Bonds	12800	3.7
EU	Deutsche Börse	Equities	30	1.1
		Bonds	1174	0.3
	Euro Stoxx 50	Equities	50	2.8
		Bonds	4554	1.1

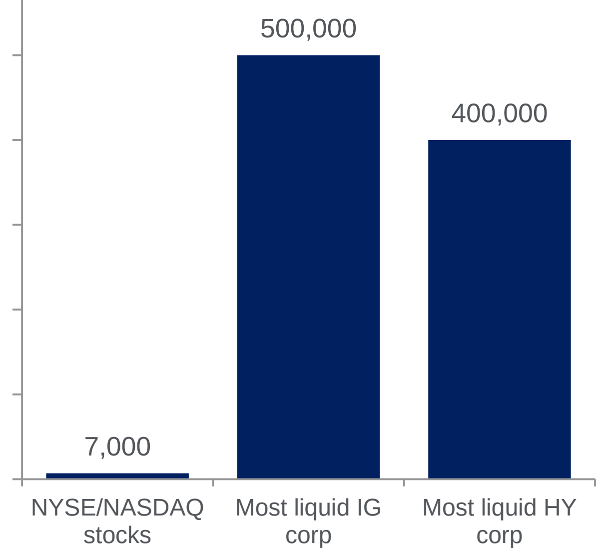
Source: Bloomberg, May 2015.

Bonds trade less often…
Average # trades per day



Source: McKinsey, NASDAQ OMX Group, NYSE Euronext, FINRA TRACE. Uses top 50 bond issues by trading frequency and value traded in 2012.

…but in much larger sizes
Average trade size, $



Source: McKinsey, NASDAQ OMX Group, NYSE Euronext, FINRA TRACE. Uses top 50 bond issues by trading frequency and value traded in 2012.

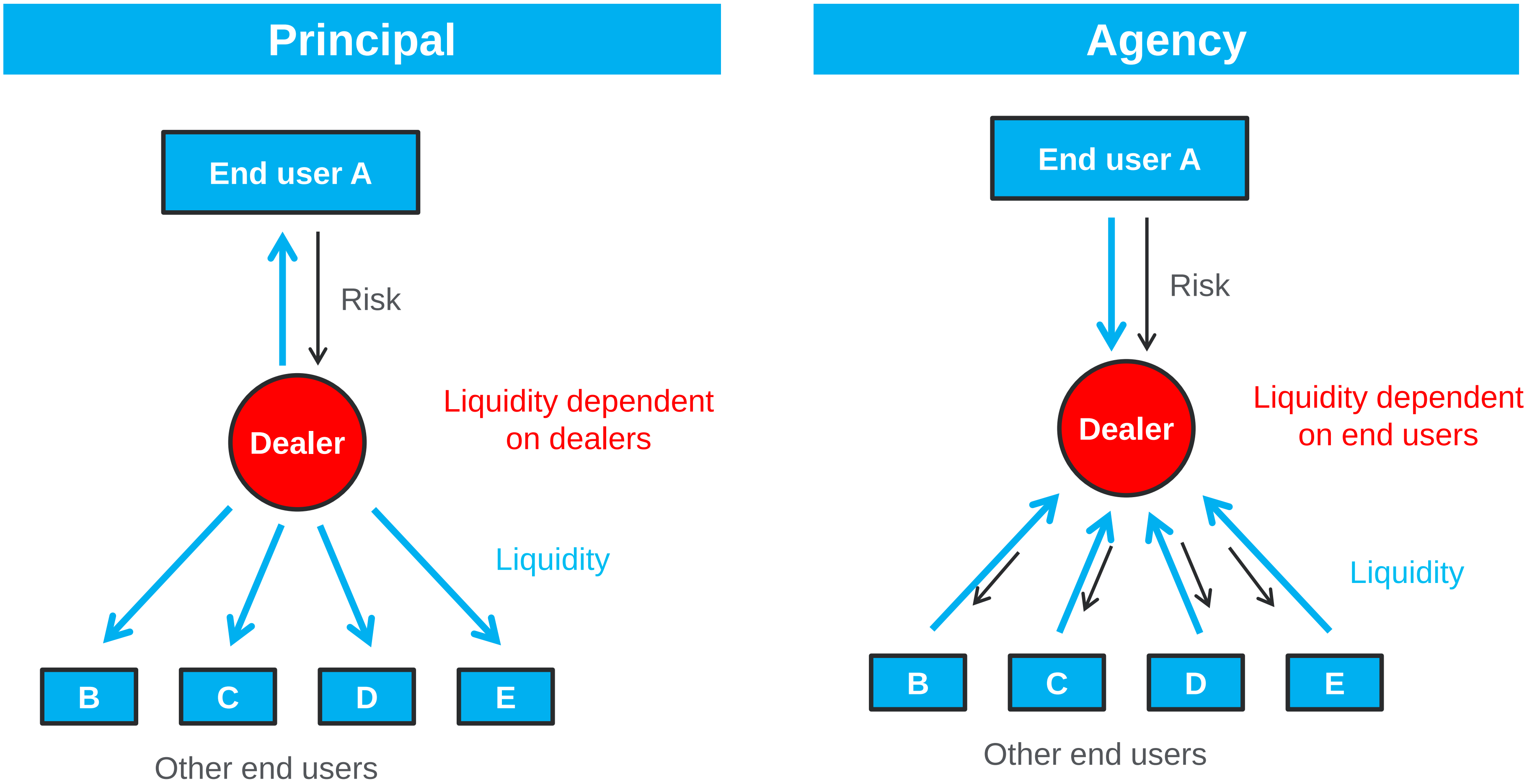
In fixed income, few trades are spread out across many bonds

Principal- versus agency-driven markets (1)

Principal	Agency
• Typically non-standardized products	• Typically standardized products
• Larger average ticket sizes; less retail involvement due to instrument features	• Smaller average ticket sizes; strong retail involvement due to simplicity of instruments
• No easy borrow, particularly on small size	• Easy borrow facilitates two-way flow
• Little or no algorithmic market-making	• Algorithms promote ticket proliferation
• Sustained two-way flow unlikely without market-makers to commit discretionary capital	• Sustained two-way flow likely without market-makers to commit discretionary capital
• Liquidity takers less willing/able to manage their own execution risk; rely on discretionary capital to warehouse or move risk	• Liquidity takers more willing/able to manage their own execution risk by breaking trades into smaller lots in firm orders
• Counterparty identity more important due to lack of active daily trading in most instruments	• Counterparty identity less important due to depth of liquidity
• Little, if any, firm liquidity; quote-driven trading	• Firm, disparate liquidity on same instrument across multiple venues

This is why fixed income has traditionally relied on a warehouser

Principal- vs agency-driven markets (2)



Source: PIMCO, Citi Research.

Dealer constraints mean FI now becoming less principal- and slightly more agent-driven

Agenda

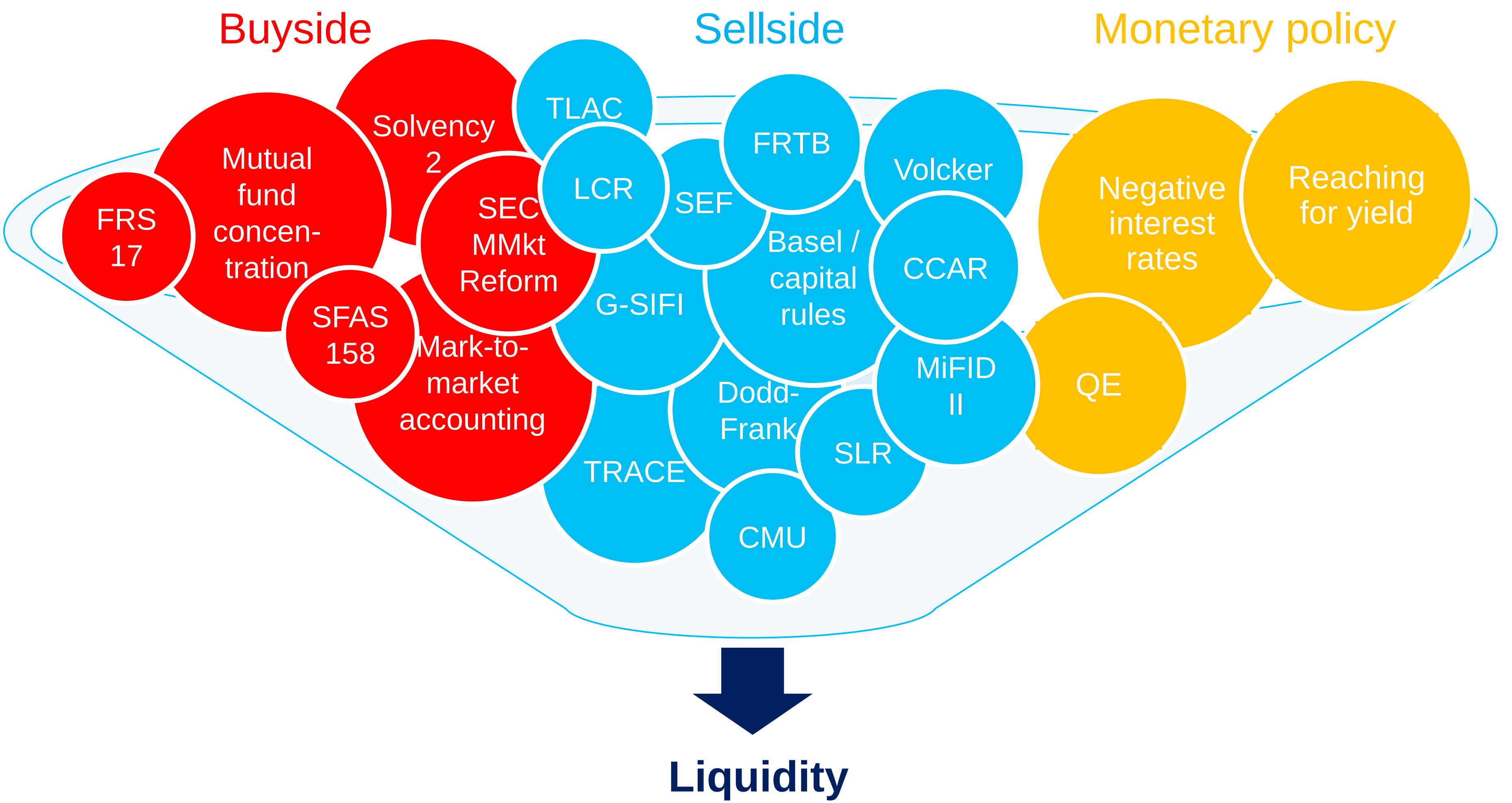
Assessing fixed-income liquidity

Regulation and structural change

Liquidity in financial shocks

Best practices and innovation

Cumulative impact of regulation and policy is difficult to isolate



But the combined effect is reduced buffers and a loss of heterogeneity

Capital rules and the impact on market-making

The role of market-maker

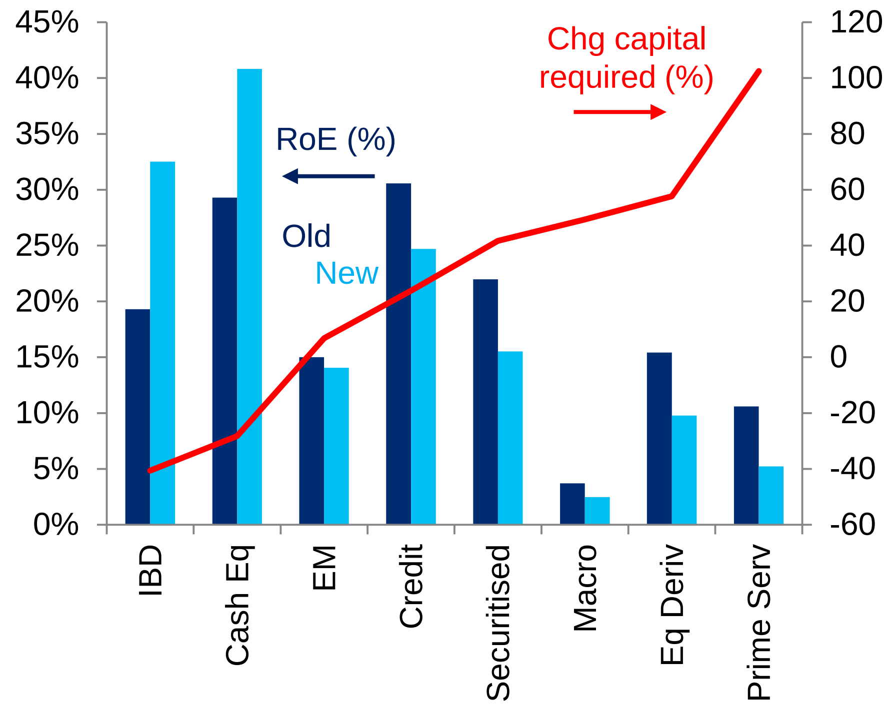
Absorbs supply and demand imbalances by transforming risk

Risk transformation

- **Curve risk (steepeners)**
- **Liquidity (off-the-run vs. on-the run)**
- **Instrument type (funded vs. unfunded)**
- **Replication (Index reconstruction)**

Capital costs have risen

Bank* RoE and change in capital estimates under old and new leverage ratio constraints, by business, %

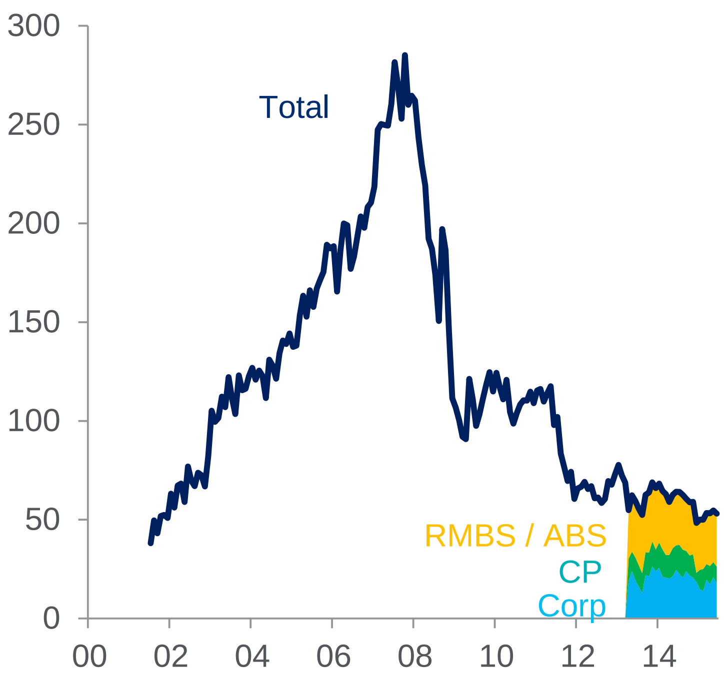


Source: Citi Research. *Numbers derived from one particular bank but are likely to be representative.

Market-making has become more expensive in the many cases where trading requires risk transformation

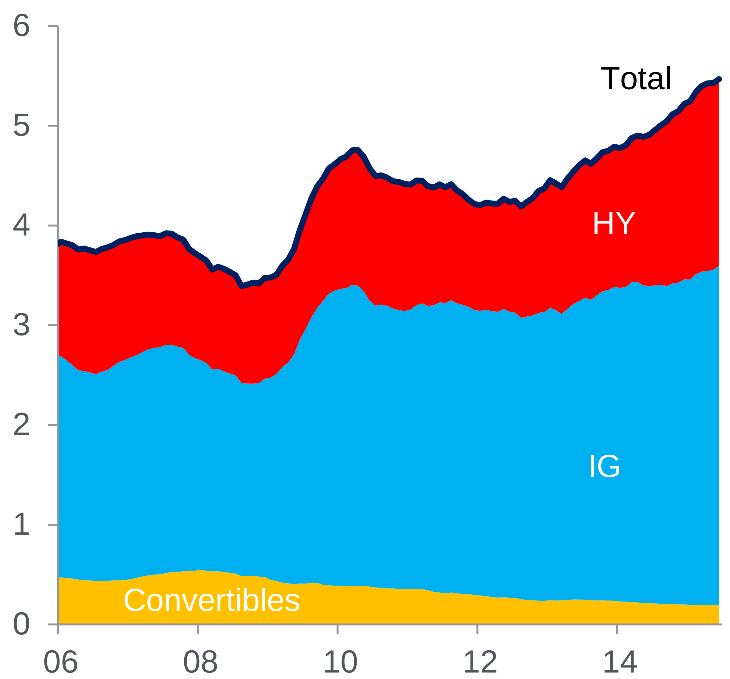
Street balance sheet has shrunk significantly

Credit* inventory down by 80%
Street holdings of credit*, $bn



Source: Federal Reserve, TRACE.
*Inventory numbers also include RMBS/ABS, which were likely much larger in 06-07.

Yet traded volumes have risen
Trading in US cash credit, rolling 12m, $tn



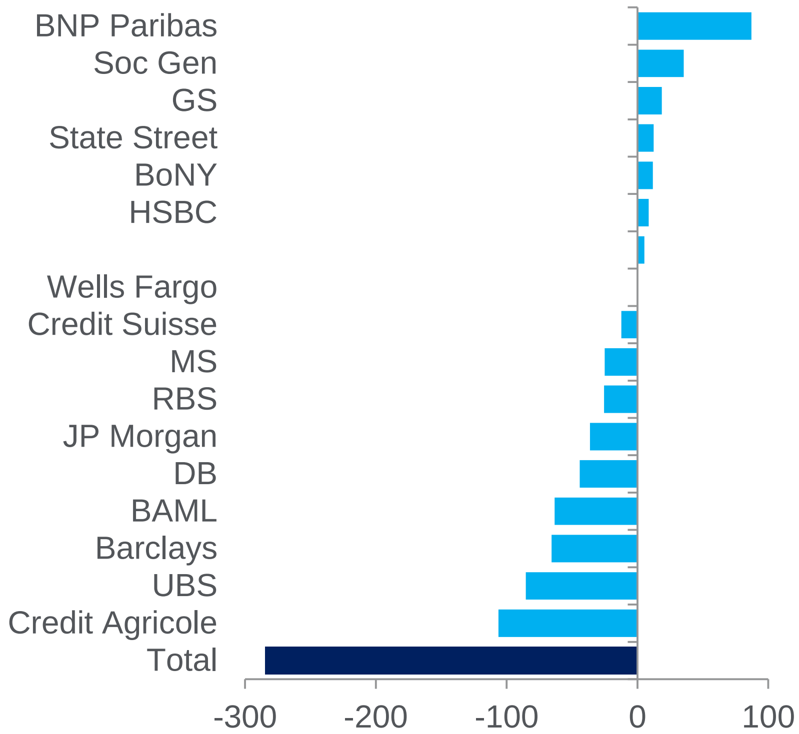
Source: TRACE.

A smaller buffer, but also one less procyclical seller in a downturn

Less leverage means lower repo

Dealer balance sheet contraction…

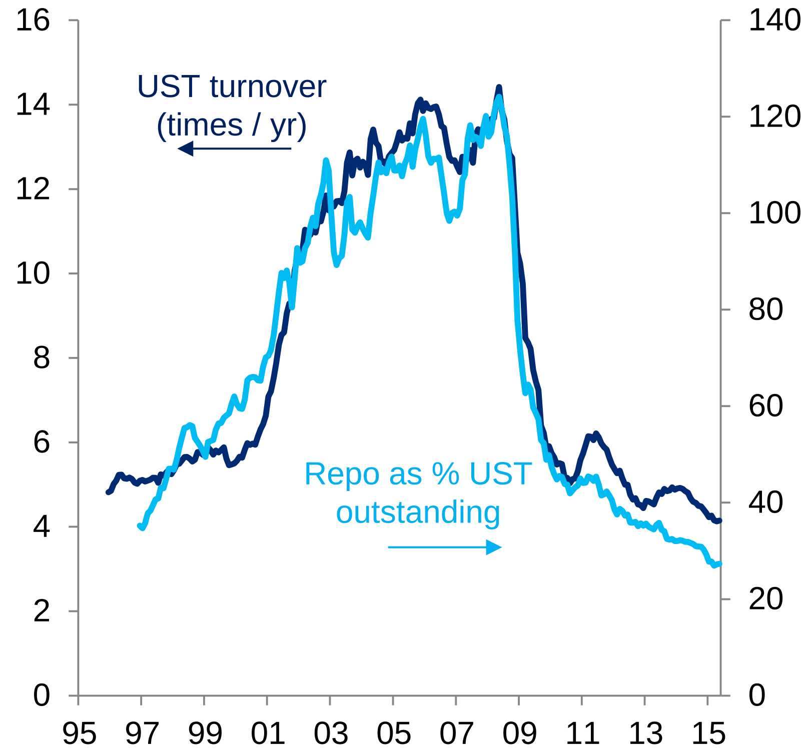
Change in repo book size, 3Q14 vs 4Q12



Source: Company accounts.

…directly impacts traded volumes

UST turnover and dealer reverse repo, vs outstanding



Source: NY Fed.

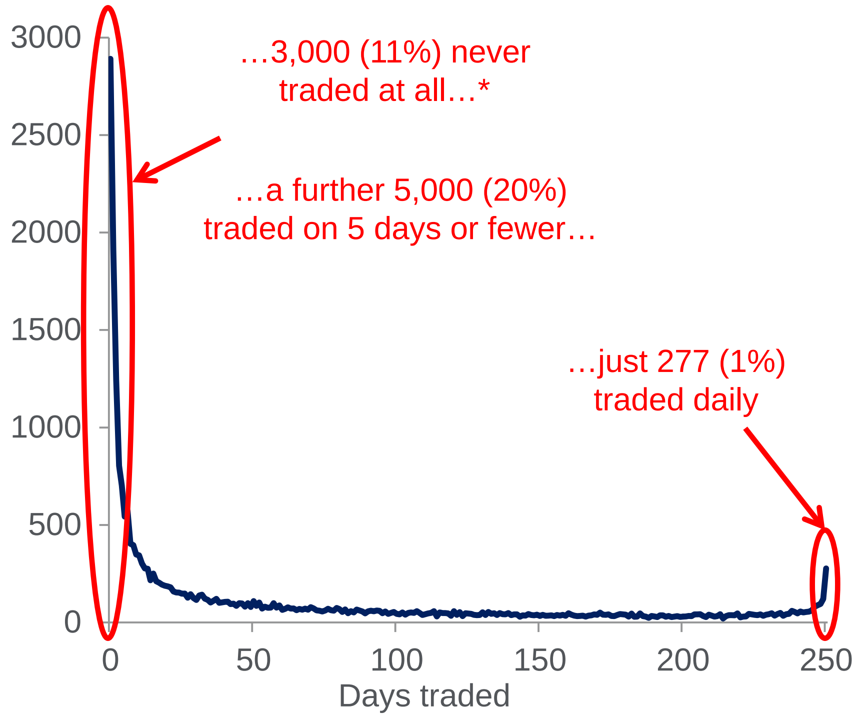
Reduced relative value trading, especially in low-risk markets

Greater transparency does not always equal greater liquidity

The liquid core is very, very small…

Trading days for US corporate bonds, FY 14

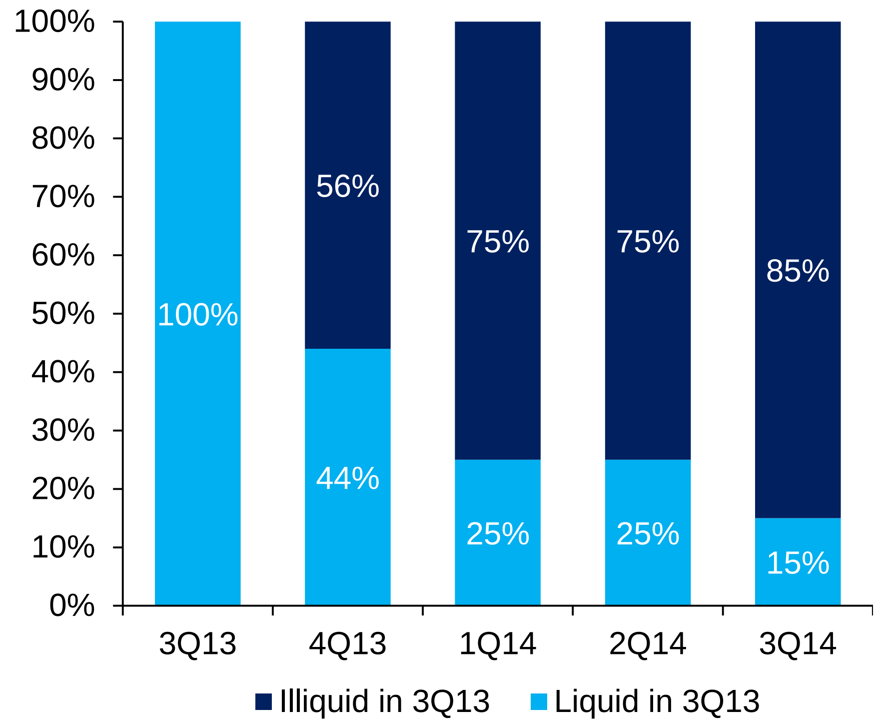
Of 26,000 publicly registered bonds outstanding last year…



Source: Citi Research, MarketAxess, Bloomberg, TRACE.
*This is just the number of publicly registered bonds which didn't trade. The total number – including private placements and the like – is likely far higher.

…and it migrates through time

Proportion of bonds deemed liquid* in 3Q13 also liquid* in subsequent quarters, € corporates



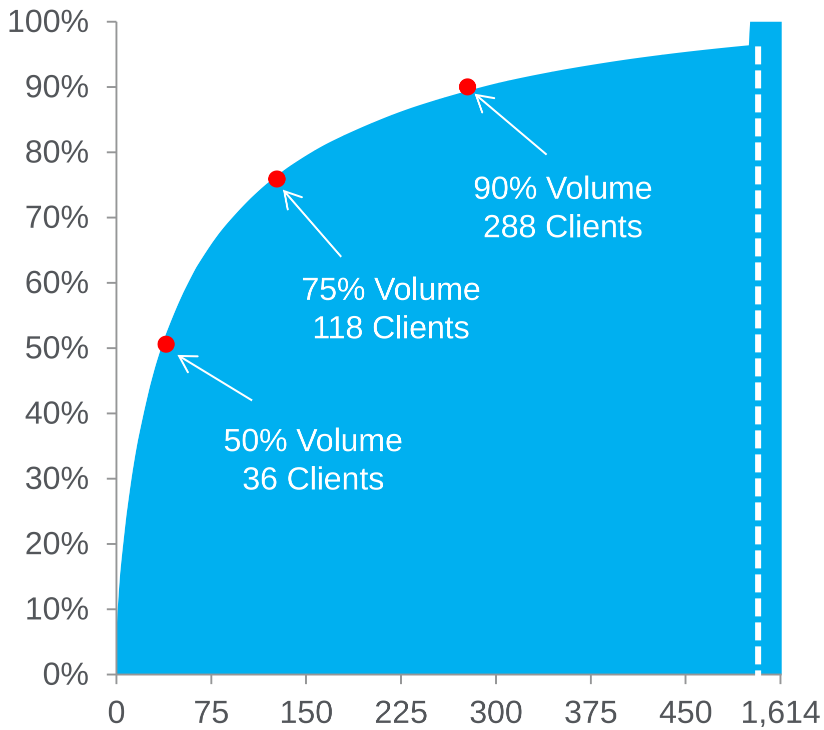
*Uses ESMA Scenario 6 definition of liquidity: >2,400 trades per year, average daily volume of at least €1m, plus requirements on the number of days the bond traded and number of participants in the market.

Source: MarketAxess Research, Trax.

Helpful when properly calibrated, but one size does not fit all

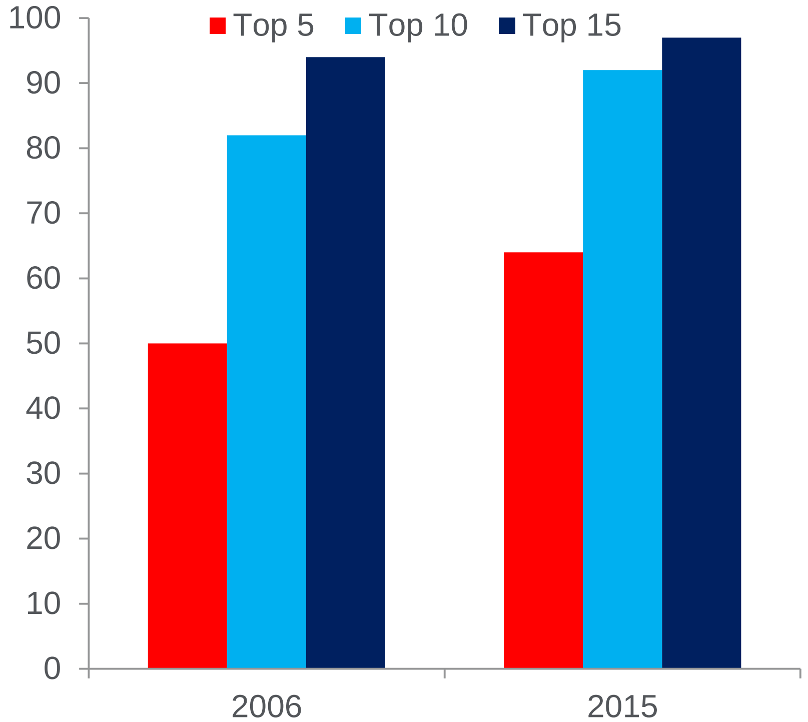
Concentration exacerbates the issue

Half the volume from just a few investors…
% total US IG volume of a large dealer, FY 2014



Source: Dealer trade data.

…who themselves are using fewer dealers
Proportion of investor trading by number of dealers, US IG, %

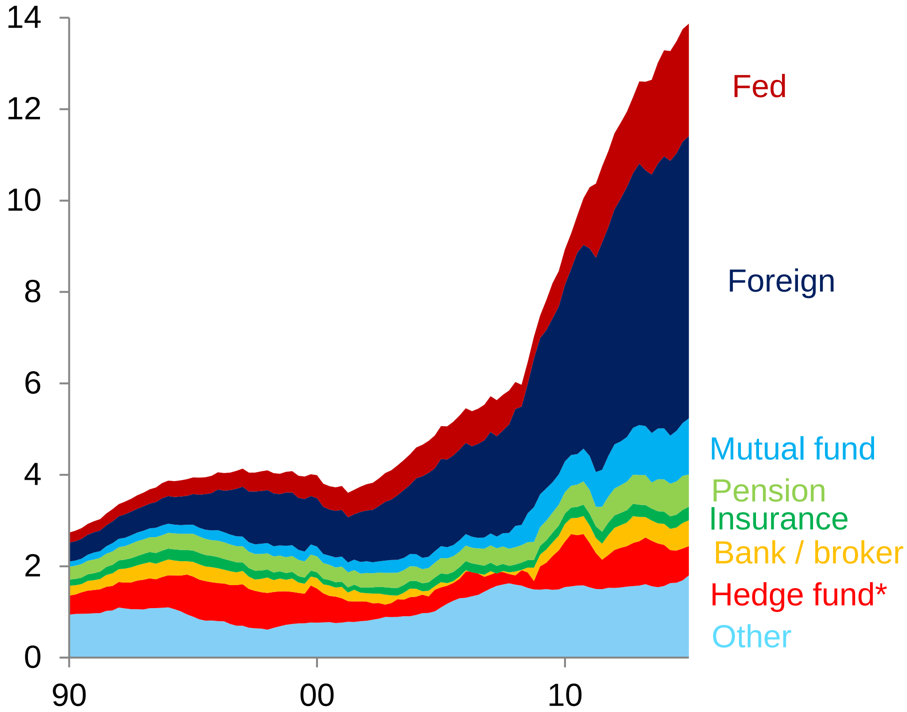


Source: NAM Greenwich Associates Survey.

Illiquidity more likely when activity dominated by a few participants with similar strategies

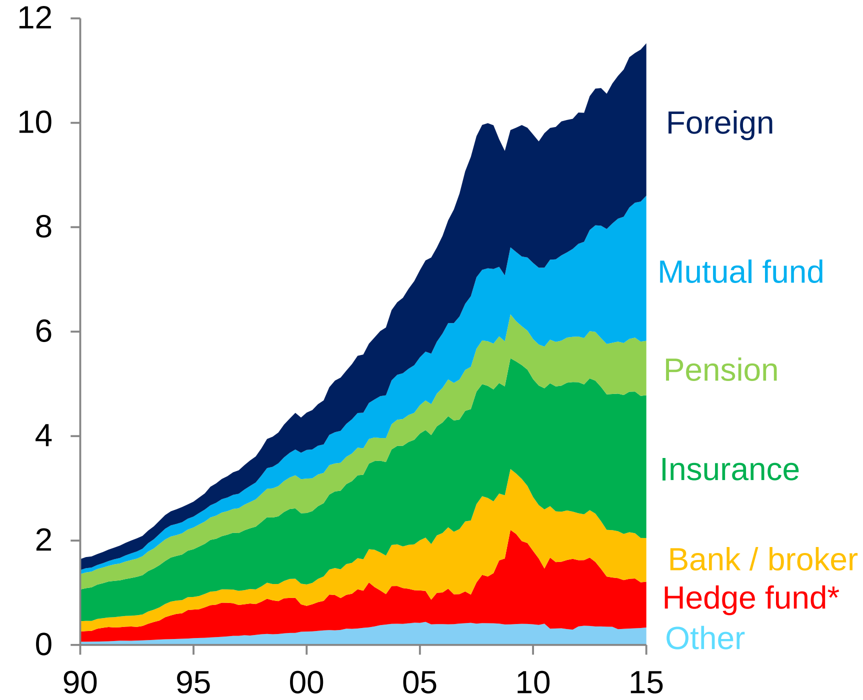
As do shifts in market structure

Who owns US Treasuries?
Outstandings by owner, $tn



Source: Federal Reserve Flow of Funds, Haver Analytics.
*Although described as “household” in flow of funds, series likely dominated by hedge funds.

Who owns US corporates?
Outstandings by owner, $tn



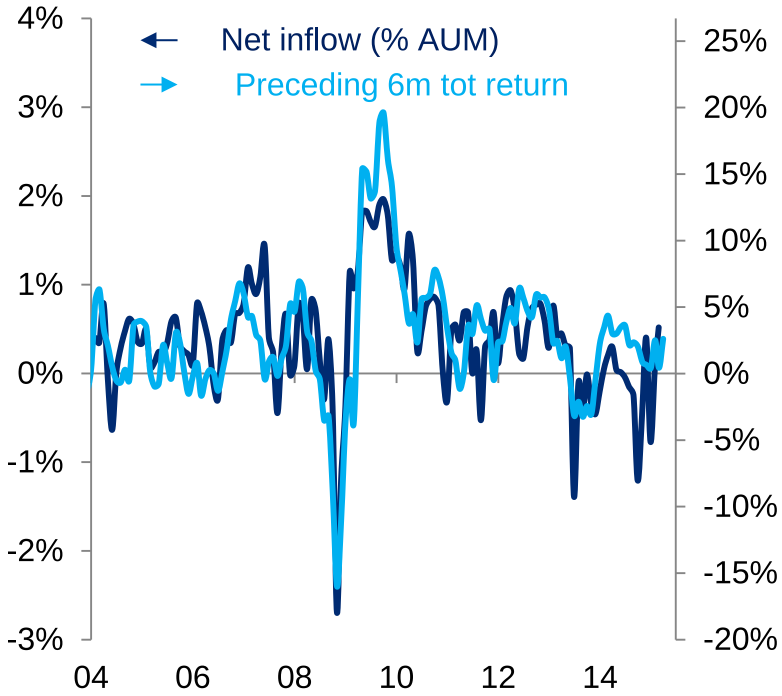
Source: Federal Reserve Flow of Funds, Haver Analytics.
*Although described as “household” in flow of funds, series likely dominated by hedge funds.

Ownership, and especially net buying, has become much less heterogeneous

Especially when so much buying is procyclical

Mutual fund investors chase returns

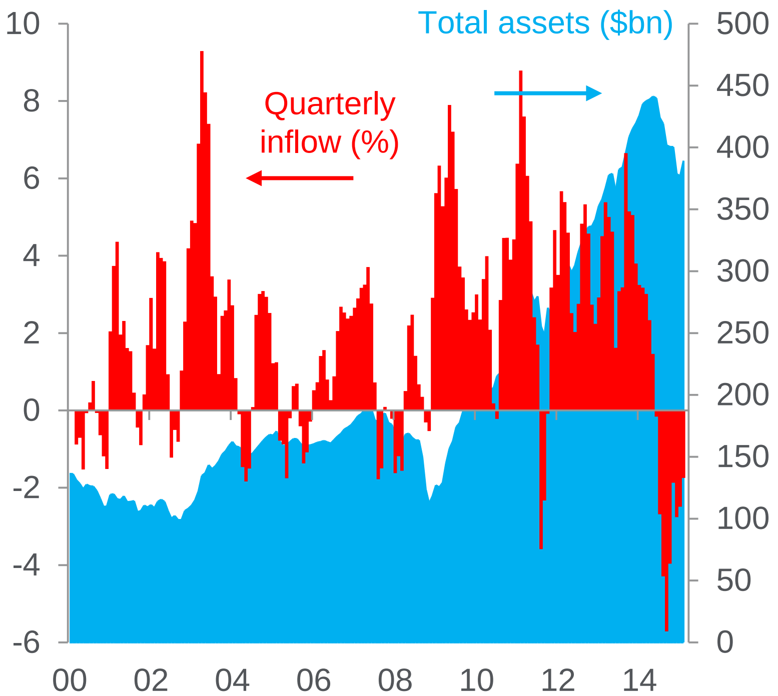
US IG+HY fund inflows vs prior 6m returns, % AUM



Source: ICI, Citi Research.

Higher risk of outflows

US HY fund inflows vs AUM, %



Source: ICI.

More procyclicality,
and more concentration on immediacy

Investors need more conviction to trade

The Cost of Business

Citi US Broad Investment Grade Corporate Index
2005 vs. 2015

	Then	*Now*
Transaction Cost Scenario	**7/21/2005**	**7/21/2015**
Citi USBIG Corporates Index YTM	**5.19%**	**3.40%**
Bid-Ask	**2-3 bps**	**5-10 bps**
Assume: Midpoint of Bid-Ask	*2.5 bps*	*7.5 bps*
Duration	5.9	7.0
Bid-Ask Cost, adjusted for duration, round trip	22 bps	79 bps
Transaction Cost as % of Par	**0.22%**	**0.79%**

USBIG Corporates Index Snapshot	**7/21/2005**	**7/21/2015**
Eff Duration, USBIG, Corp.	5.9	7.0
Spread to Treasury, USBIG, Corp.	+100 bps	+143 bps
# of Issues, USBIG, Corp.	2,671	5,619
Yield to Maturity, USBIG, Corp.	5.19%	3.40%
Market Value, USBIG, Corp. ($ bn)	$1,680	$4,394
Average Life, USBIG, Corp. (years)	9.3	10.5

Source: Yield Book.

Suppose You Had $10 Million….

Expected Transaction Costs for Switch Trade, 1 Year Horizon
Holding Spreads & Yields Constant (No Roll-Down)

	Then	*Now*
Expected Transaction Costs	**7/21/2005**	**7/21/2015**
Yield Return, Gross	$519,053	$339,824
Bid-Ask Cost, adjusted for duration, round trip	$21,989	$78,704
Transaction Cost as % of Yield	**4.24%**	**23.16%**

Consider the Break-Even Horizon…

Using Yields & Transaction Costs in 2005 & 2015
Holding Spreads & Yields Constant (No Roll Down)

***Then* | 2005 break-even period = 15 days**

Transaction Cost = 0.22% of par; Market Yield = 5.19%

***Now* | 2015 break-even period = 85 days**

Transaction Cost = 0.79% of par; Market Yield = 3.40%

Note: This exercise assumes a switch into a similar bond (price, yield, maturity) assuming selling at mid-market and buying new security at offer side. Costs could of course be reduced by buying in primary – which may be one reason activity has migrated in that direction.

Higher transaction costs, and lower yields and volatility: a self-reinforcing disincentive to trade

Agenda

Assessing fixed-income liquidity

Regulation and structural change

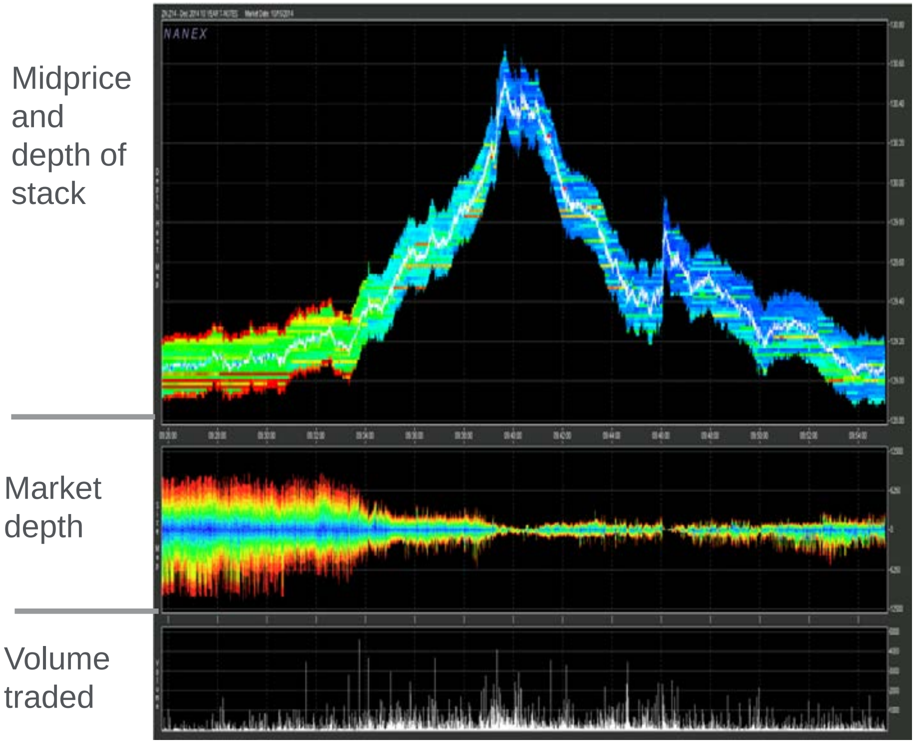
Liquidity in financial shocks

Best practices and innovation

What to make of the flash crash?

Depth just disappeared…

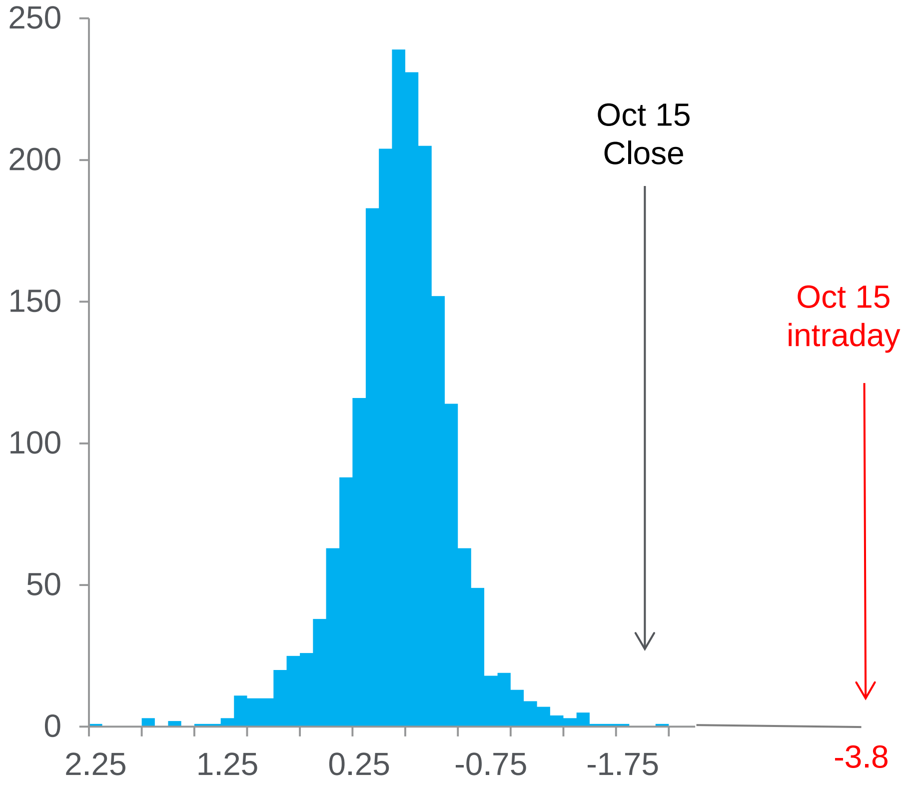
Intraday movements in 10y UST futures on 15 Oct 2014; red=better liquidity, blue=worse



Source: Nanex representation of TY data (CBOT)

…making for unprecedented price moves

Histogram of Blue Eurodollar futures daily moves expressed in standard deviations of 3m range, 2006-14



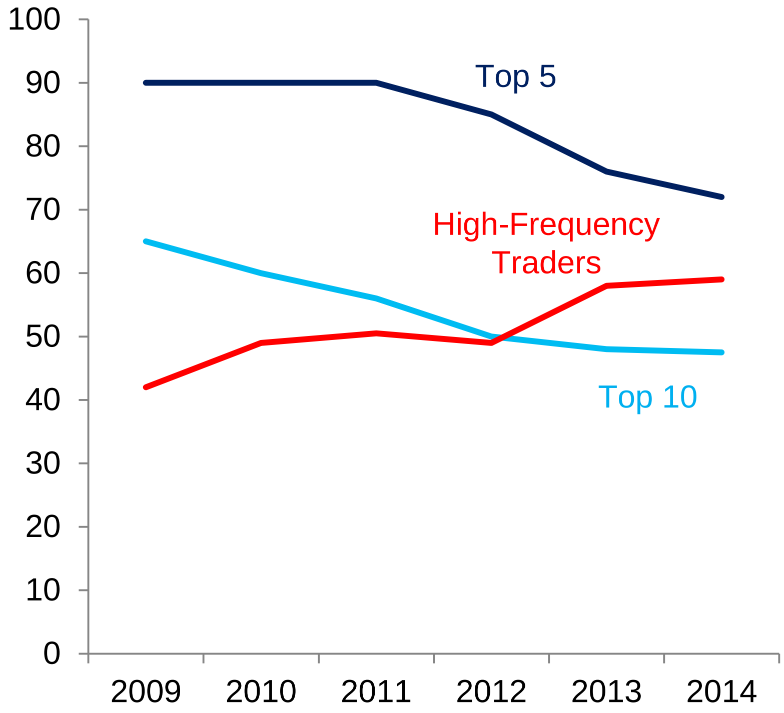
Source: CME, Citi Research.

Unusual – but hardly a once per 3 billion year event

Were there any warning signs?

Increased HFT participation, perhaps

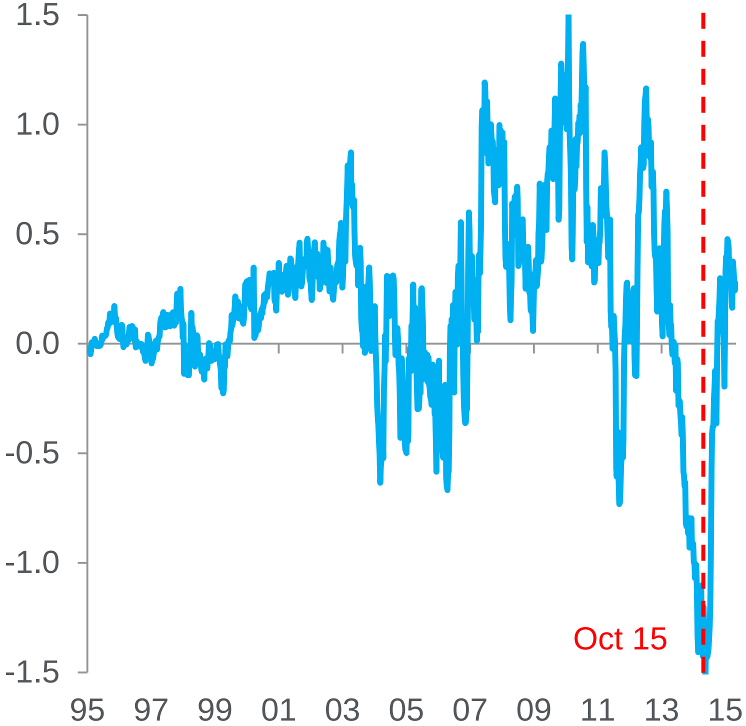
Proportion of inter-dealer trading by counterparty in USTs, %



Source: Brokertec.

Positioning was also extreme

Net speculative positioning in Eurodollars, million contracts



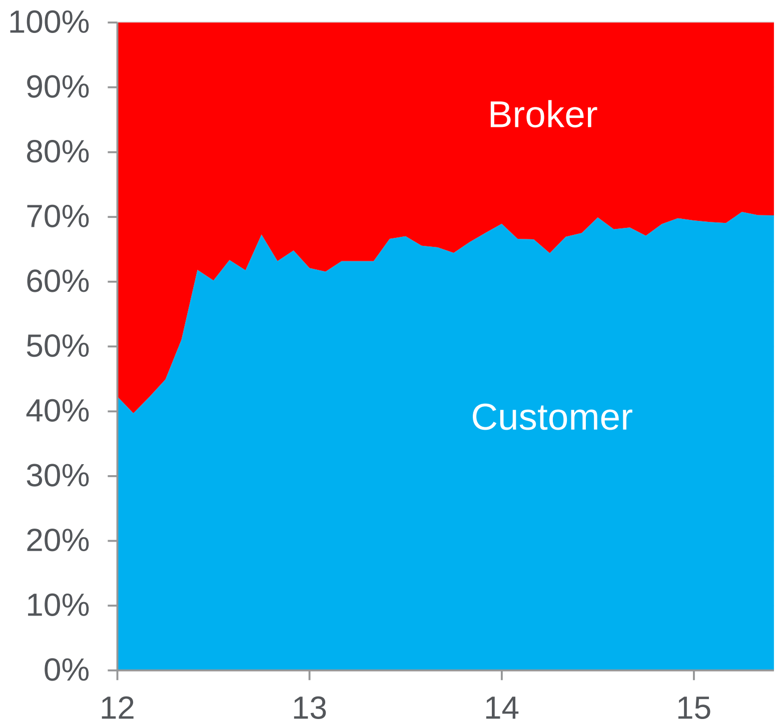
Source: CFTC.

Dealers and HFTs suddenly pulled away

But IDB trades are only a fraction of the market

Investors choose dealers over brokers…

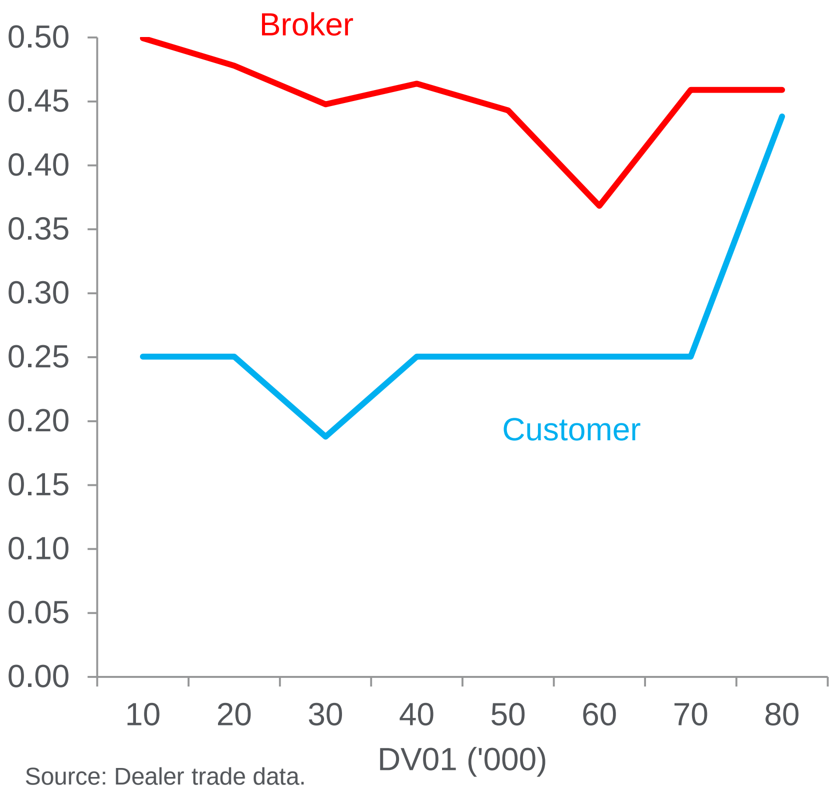
Sample dealer, UST trading by counterparty



Source: Dealer trade data.

…in part because of better bid-offer

Execution cost by trade size and counterparty, UST OTR 7-30y, distance from mid (ticks)



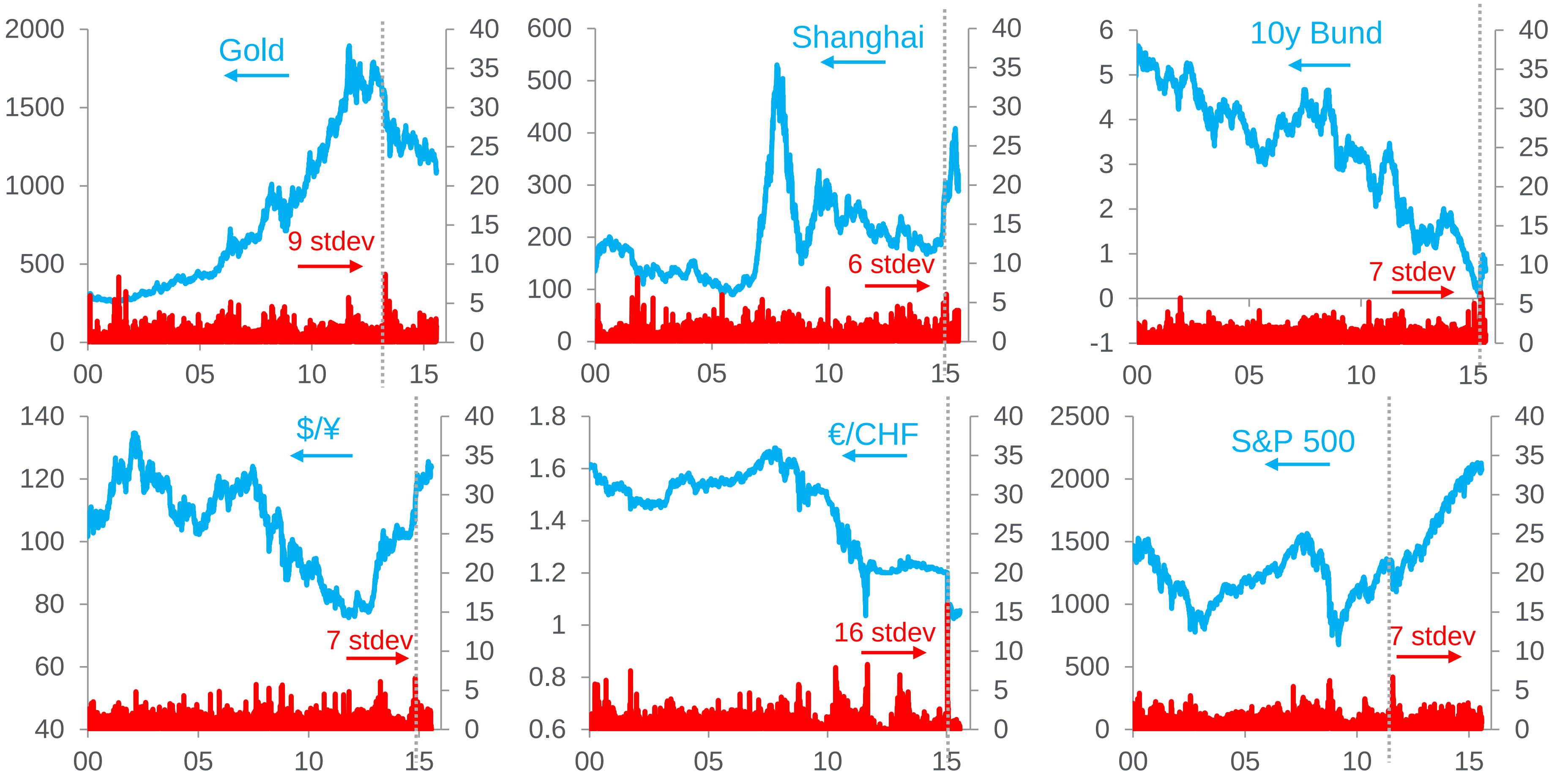
Source: Dealer trade data.

Focus on inter-dealer trades exaggerates the importance of HFT

Other markets are suffering bouts of illiquidity too

Just a widespread case of the hiccups?

Daily absolute price moves expressed as multiple of 1y standard deviation

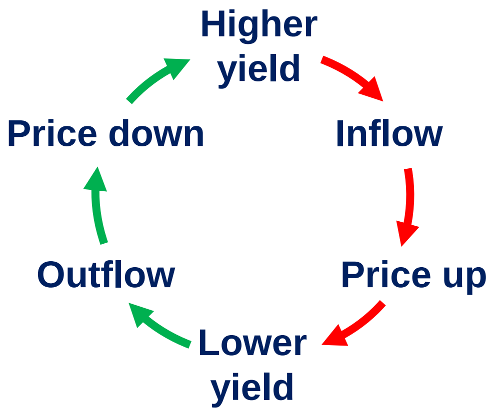


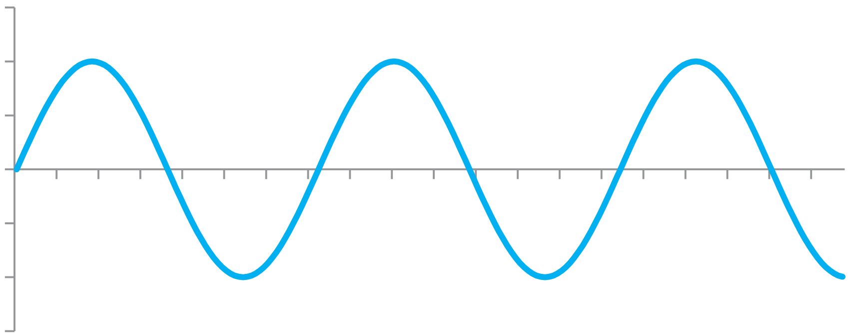
Source: Haver Analytics.

This can't just be about dealer balance sheet

Have market dynamics changed?

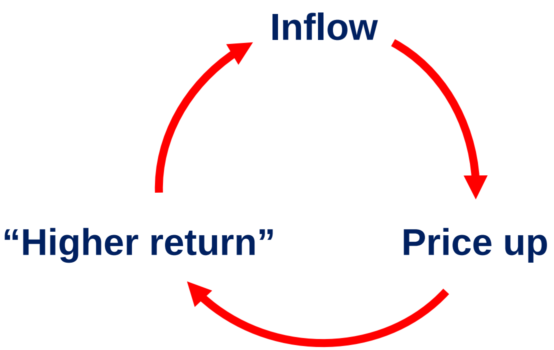
Normally, price movements are self-limiting
Negative feedback loop

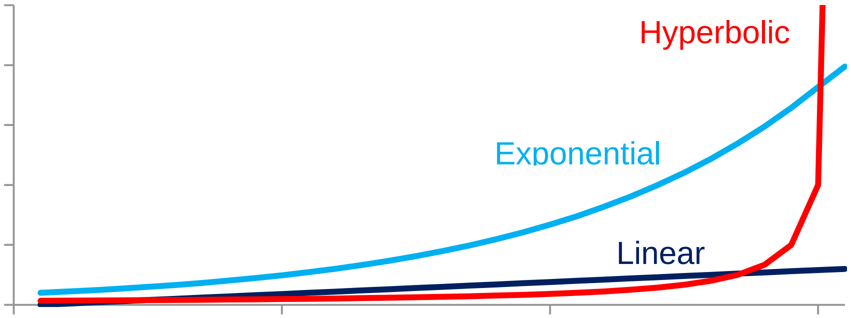




Source: Citi Research.

Today's moves feel different
Positive feedback loop





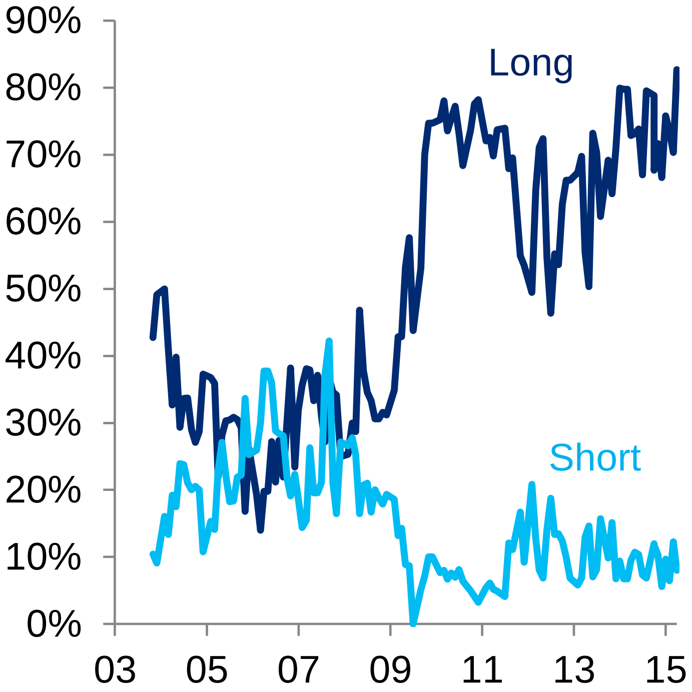
Source: Citi Research.

Investors buying not on value but on inflows

Evidence of herding

One-way market

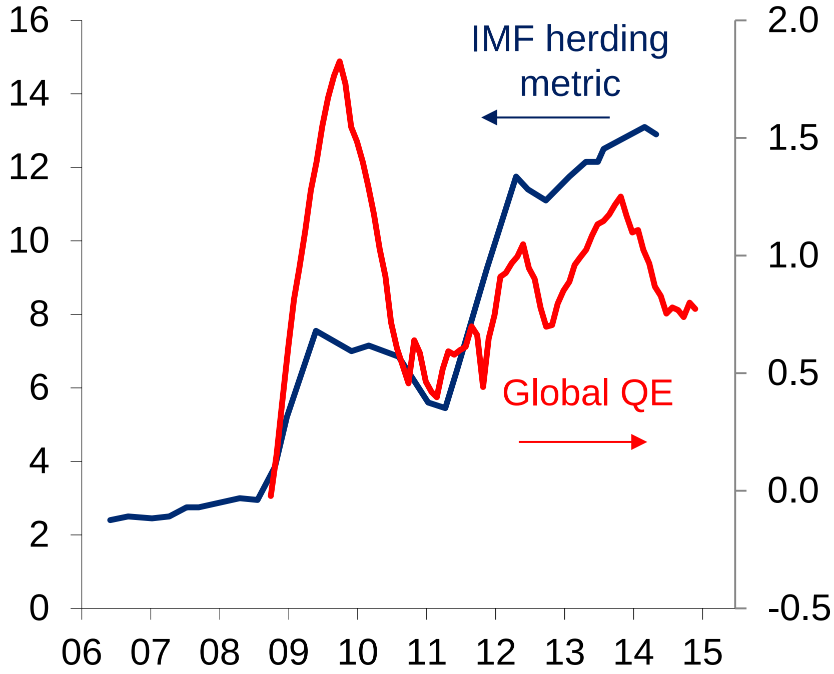
Dealer credit survey positioning, % of respondents



Source: Citi Research.

More QE, more crowded positioning

Herding index* vs global central bank purchases, rolling 6m, $tn



Source: IMF, Citi Research.

* Derived from correlation between security-by-security transactions in US mutual funds across a variety of asset classes.

Extreme monetary policy has caused herding - removing it will be difficult

Agenda

Assessing fixed-income liquidity

Regulation and structural change

Liquidity in financial shocks

Best practices and innovation

What should be done?

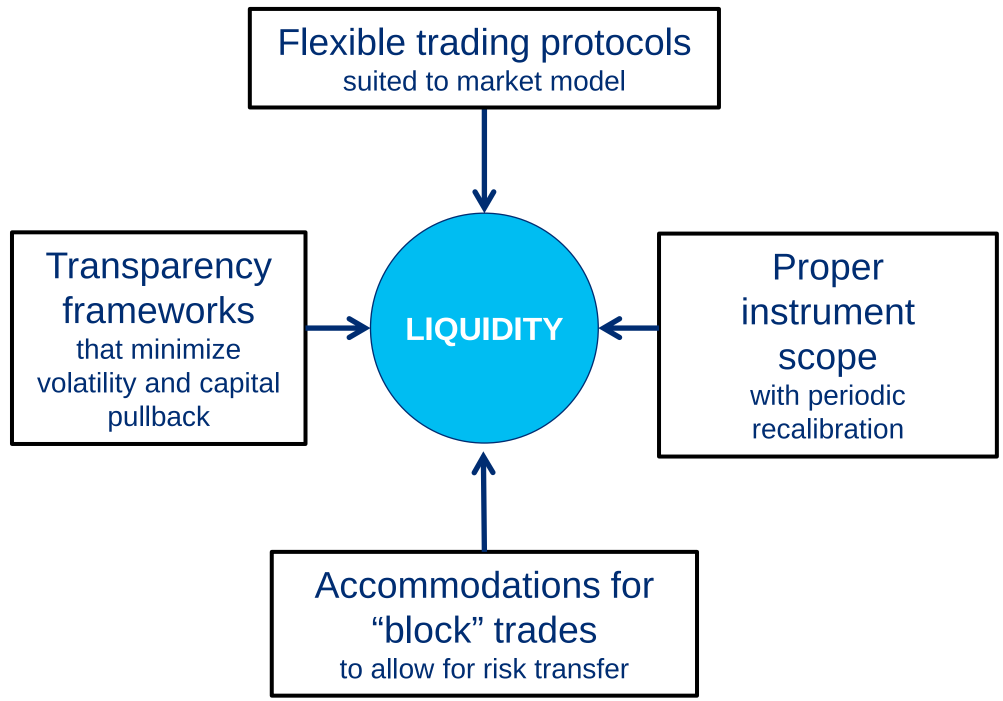
Action and reaction

Potential measures and complications

End user repo clearing	• Risk sharing problematic
Regulate end users	• Hard to enforce ‘skin in the game’
Regulate venues	• Sensible, but will not eliminate vol
Encourage all-to-all trading	• Works only when 2-way interest exists
Greater transparency and standardization	• Requires careful calibration by market

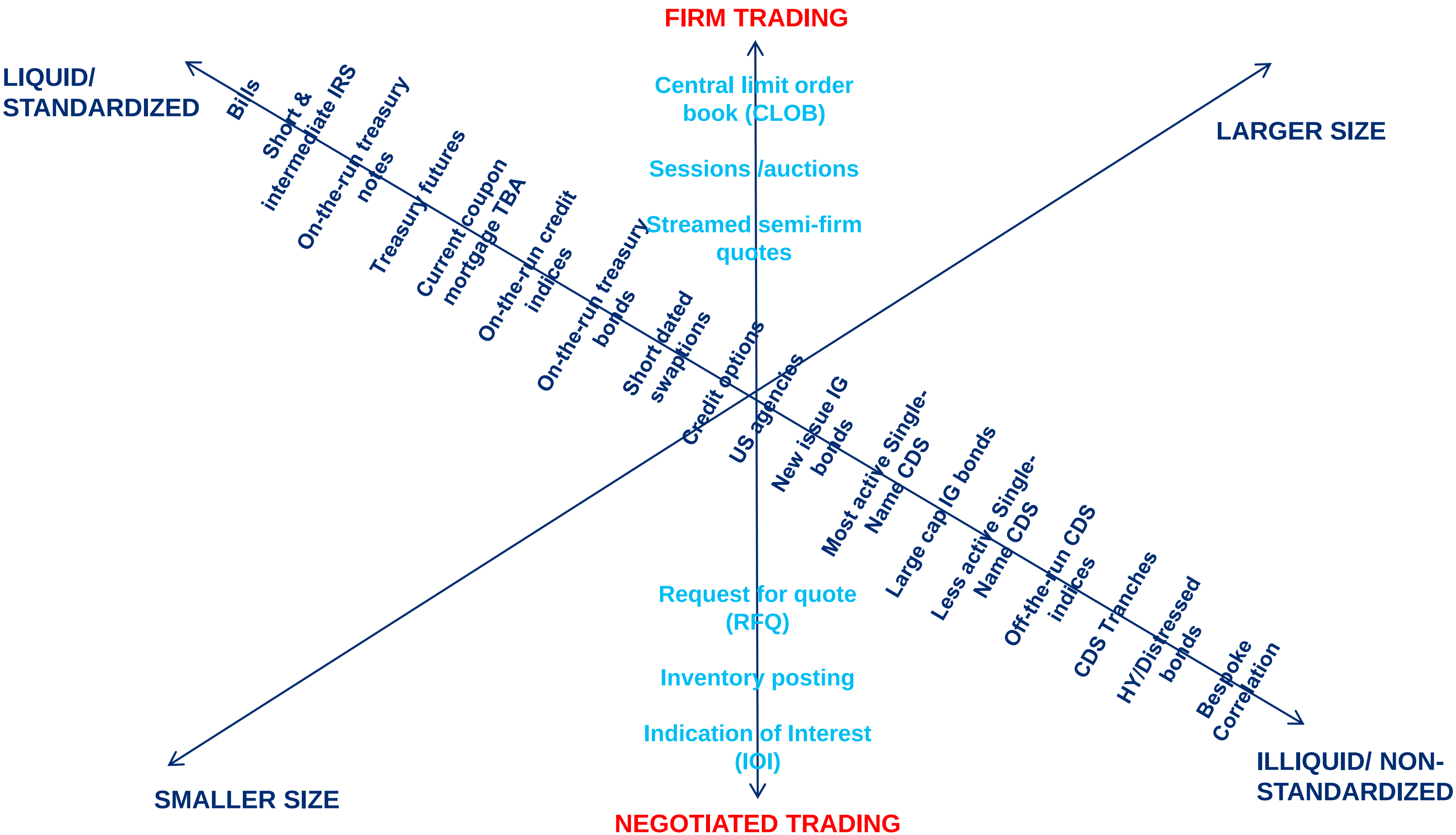
Balance evolution with revolution

Use a phased approach, and consider multiple dimensions



Multiple measures possible – none of which will eliminate the problem entirely

Protocols should accommodate market and allow evolution



Think less of enforcing participation, and more about removing barriers to confidence

Building safe electronic markets

SEC Chair, Mary Jo White

Commentary to the major equities markets following the 08/01/12 Knight Capital Market-Making software glitch and the 08/22/12 NASDAQ trade halt

The relevant parties in consultation with other market participants will:

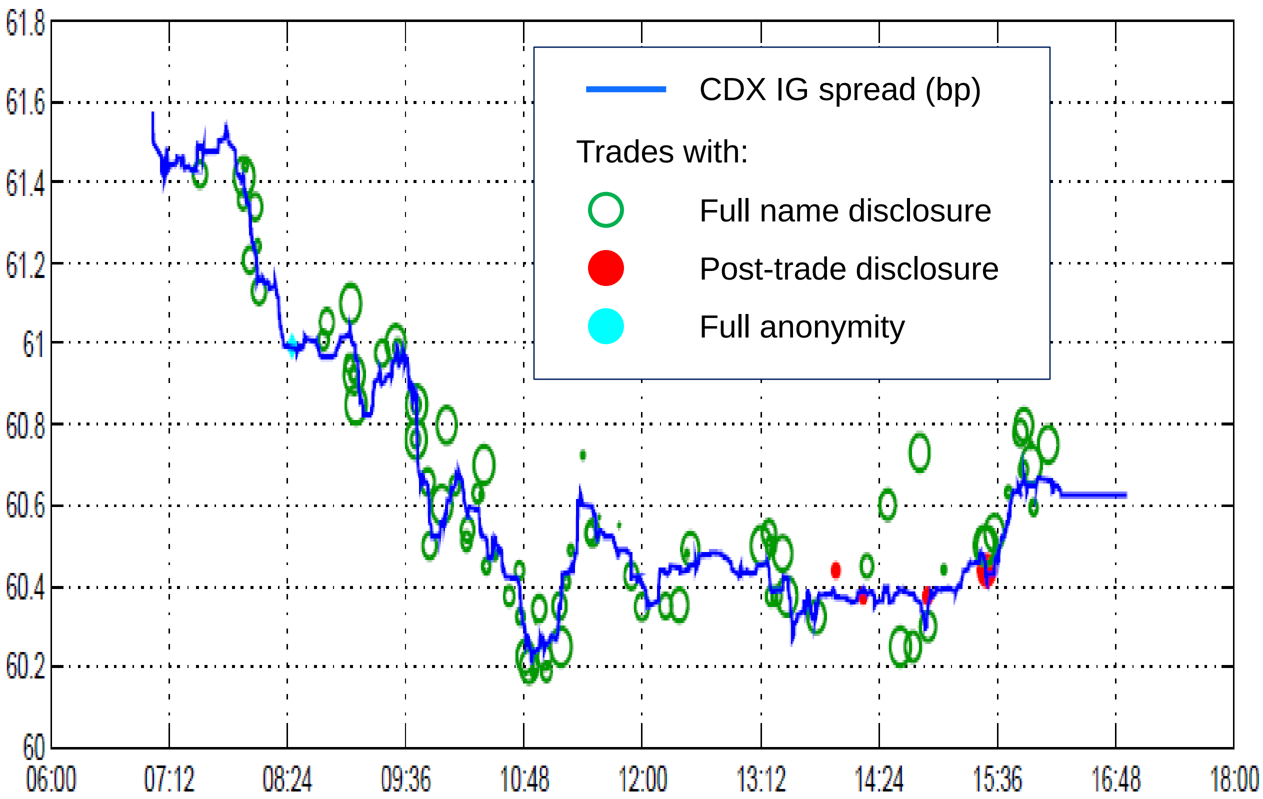
- Provide comprehensive action plans that address the standards necessary to establish highly resilient and robust systems for the securities information processors (SIPs), including testing standards and disclosure protocols.
- Identify and provide assessments of the robustness and resilience of other critical infrastructure systems.
- Provide SIP plan and/or rule amendments addressing the issuance, effectiveness, and communication of regulatory halts.
- Review their rules relating to the trade break process and procedures to reopen trading following a trading halt, and provide amendments to those rules as necessary.
- Provide rule amendments to implement “kill switches” that would allow exchanges to shut down trading in the event of technological failures, and review and consider other potential risk mitigation mechanisms.

Best practices from more evolved electronic markets still missing in FI

Post-trade anonymity

Anonymity proves unpopular

CDX IG SEF trade analysis by trading venue and trade size (diameter), 15 Apr 2015



Source: Bloomberg SDR.

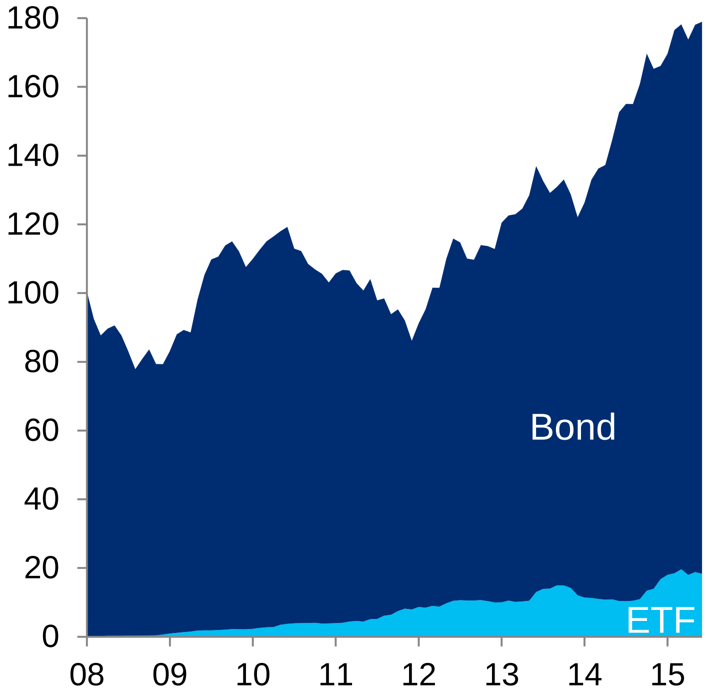
- More trades and larger sizes executed via fully name disclosed RFQ protocol
- Liquidity seekers currently prefer bilateral dealer-client platforms because of greater liquidity and higher execution certainty

Liquidity best on platforms with name give-up

ETFs

Adding to liquidity, not reducing it

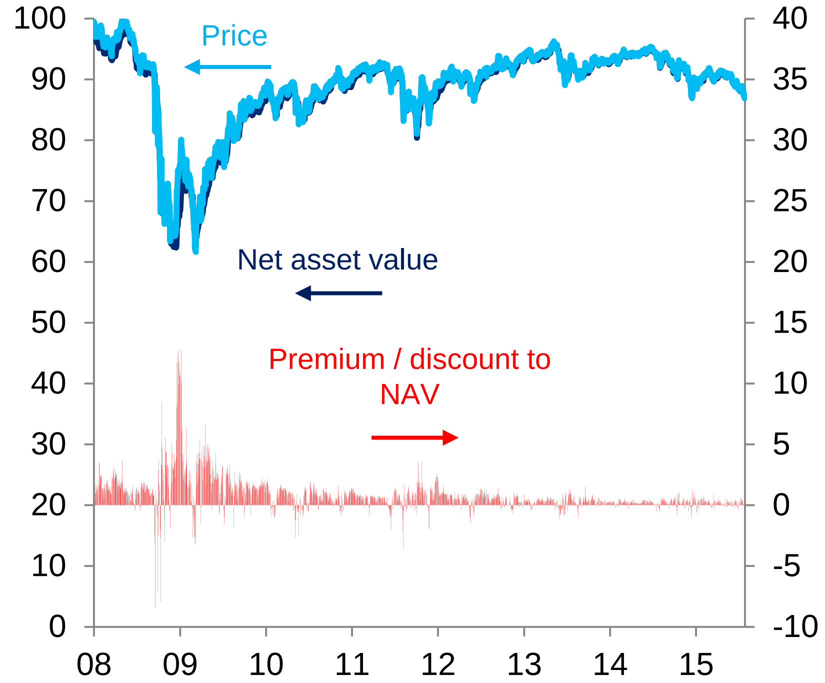
HY cash trading by instrument, 6m avg, $bn



Source: Bloomberg.

A built-in buffer for shocks

HY ETF price and net asset value comparison



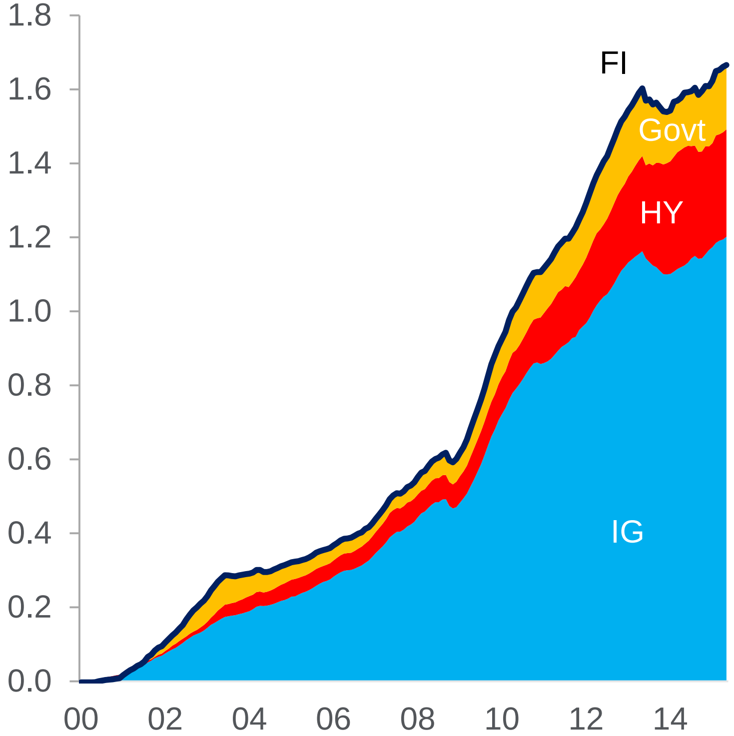
Source: Bloomberg.

Ability to trade at discount to NAV means buffer built in

Mutual fund stress tests and liquidity buffers

The industry's growth may be steady

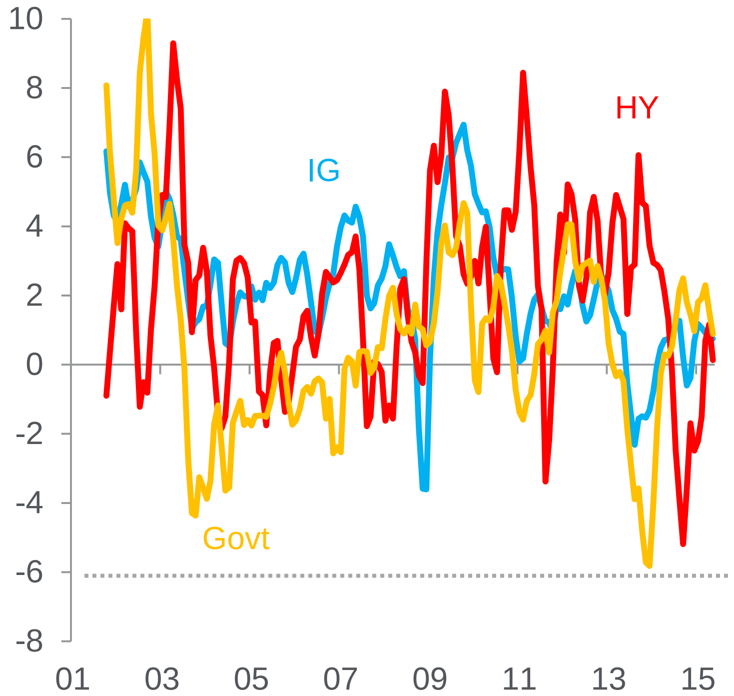
Cumulative net sales since 2000, mutual fund + ETF, $tn



Source: ICI.

But individual fund flows vary substantially

Net quarterly sales by asset class, % AUM, mutual fund + ETF



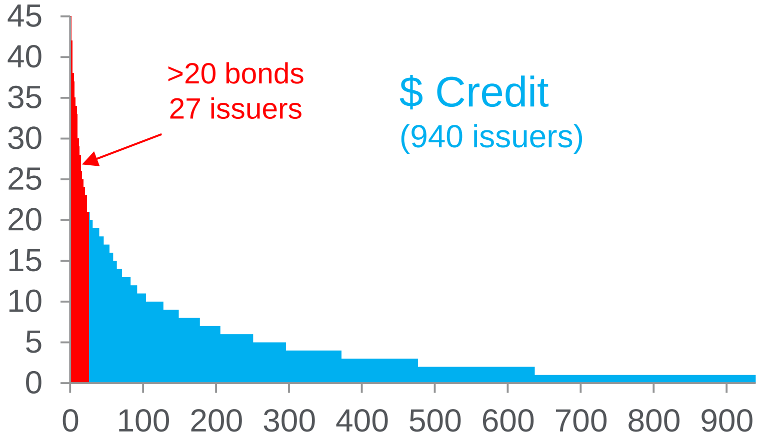
Source: ICI.

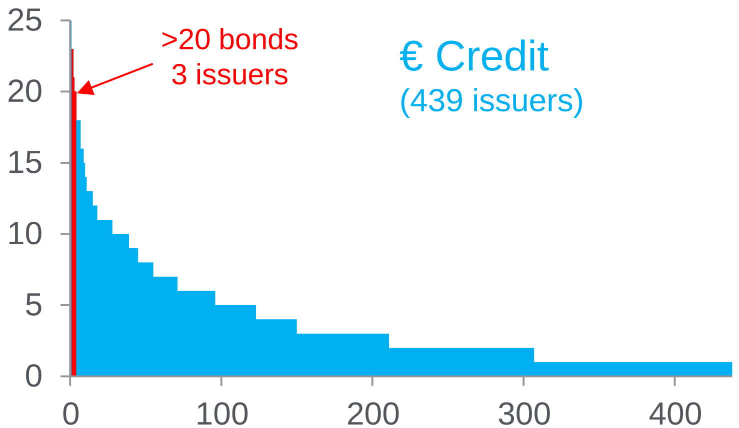
Encourage safeguards against firesales, whether self-imposed or regulated

Standardizing corporate issuance

Most issuers have only a few bonds…

Number of bonds outstanding by issuer, $ and € IG





Source: Markit.

…and cherish their flexibility

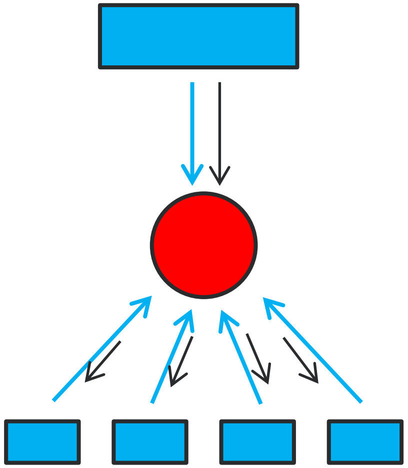
Considerations upon issuance consolidation

Maturity Management	• Large Towers can cause significant refinancing and credit risk
Reopening Bond amid Fluctuating Rates and Credit Markets	• Potential OID and Premium issues
Bespoke Transactions for Each Issuer	• Potential impediment on capital raising strategies
Standardized Coupons	• Hinder cash management strategies
Covenant Language	• Could be used by issuers, limiting bond reopening
Infrequent, Small issuers	• Elevated borrowing costs as a result of not reaching the threshold
Secondary Market Underperformance	• Reopening of bonds could push issuers wider

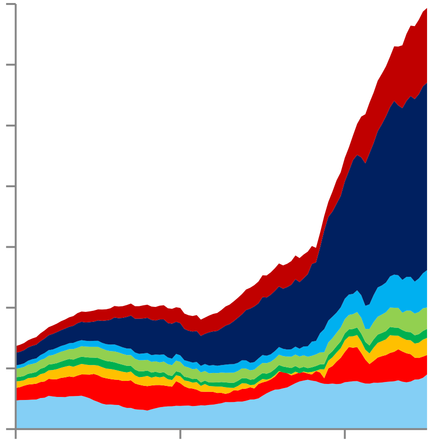
Would help only a small number of borrowers

Conclusion

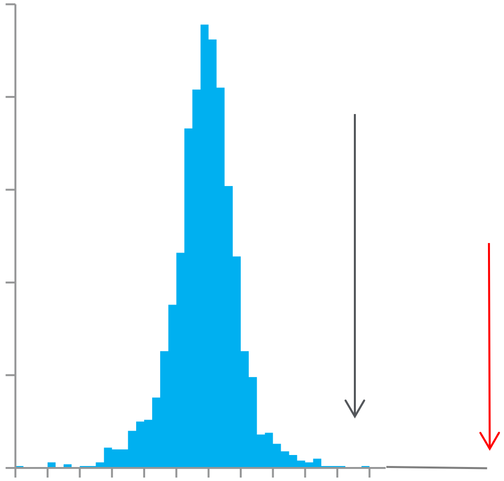
A shift towards an agent-based model



…and a loss of heterogeneity…



…leaves liquidity vulnerable to the whims of end users



Expect more bouts of illiquidity ahead

Appendix A-1

Analyst Certification

The research analyst(s) primarily responsible for the preparation and content of this research report are named in bold text in the author block at the front of the product except for those sections where an analyst's name appears in bold alongside content which is attributable to that analyst. Each of these analyst(s) certify, with respect to the section(s) of the report for which they are responsible, that the views expressed therein accurately reflect their personal views about each issuer and security referenced and were prepared in an independent manner, including with respect to Citigroup Global Markets Inc and its affiliates. No part of the research analyst's compensation was, is, or will be, directly or indirectly, related to the specific recommendation(s) or view(s) expressed by that research analyst in this report.

IMPORTANT DISCLOSURES

Analysts' compensation is determined based upon activities and services intended to benefit the investor clients of Citigroup Global Markets Inc. and its affiliates ("the Firm"). Like all Firm employees, analysts receive compensation that is impacted by overall firm profitability which includes investment banking revenues.

For important disclosures (including copies of historical disclosures) regarding the companies that are the subject of this Citi Research product ("the Product"), please contact Citi Research, 388 Greenwich Street, 28th Floor, New York, NY, 10013, Attention: Legal/Compliance [E6WYB6412478]. In addition, the same important disclosures, with the exception of the Valuation and Risk assessments and historical disclosures, are contained on the Firm's disclosure website at https://www.citivelocity.com/cvr/eppublic/citi_research_disclosures. Valuation and Risk assessments can be found in the text of the most recent research note/report regarding the subject company. Historical disclosures (for up to the past three years) will be provided upon request.

NON-US RESEARCH ANALYST DISCLOSURES

Non-US research analysts who have prepared this report (i.e., all research analysts listed below other than those identified as employed by Citigroup Global Markets Inc.) are not registered/qualified as research analysts with FINRA. Such research analysts may not be associated persons of the member organization and therefore may not be subject to the NYSE Rule 472 and NASD Rule 2711 restrictions on communications with a subject company, public appearances and trading securities held by a research analyst account. The legal entities employing the authors of this report are listed below:

Citigroup Global Markets Ltd	Matt King

OTHER DISCLOSURES

Many European regulators require that a firm must establish, implement and make available a policy for managing conflicts of interest arising as a result of publication or distribution of investment research. The policy applicable to Citi Research's Products can be found at https://www.citivelocity.com/cvr/eppublic/citi_research_disclosures.

For securities recommended in the Product in which the Firm is not a market maker, the Firm is a liquidity provider in the issuers' financial instruments and may act as principal in connection with such transactions. The Firm is a regular issuer of traded financial instruments linked to securities that may have been recommended in the Product. The Firm regularly trades in the securities of the issuer(s) discussed in the Product. The Firm may engage in securities transactions in a manner inconsistent with the Product and, with respect to securities covered by the Product, will buy or sell from customers on a principal basis.

Citigroup Global Markets India Private Limited and/or its affiliates may have, from time to time, actual or beneficial ownership of 1% or more in the debt securities of the subject issuer.

Citi Research generally disseminates its research to the Firm’s global institutional and retail clients via both proprietary (e.g., Citi Velocity and Citi Personal Wealth Management) and non-proprietary electronic distribution platforms. Certain research may be disseminated only via Citi’s proprietary distribution platforms; however such research will not contain changes to earnings forecasts, target price, investment or risk rating or investment thesis or be otherwise inconsistent with the author’s previously published research. Certain research is made available only to institutional investors to satisfy regulatory requirements. Individual Citi Research analysts may also opt to circulate published research to one or more clients by email; such email distribution is discretionary and is done only after the research has been disseminated. The level and types of services provided by Citi Research analysts to clients may vary depending on various factors such as the client’s individual preferences as to the frequency and manner of receiving communications from analysts, the client’s risk profile and investment focus and perspective (e.g. market-wide, sector specific, long term, short-term etc.), the size and scope of the overall client relationship with Citi and legal and regulatory constraints.

Pursuant to Comissão de Valores Mobiliários Rule 483, Citi is required to disclose whether a Citi related company or business has a commercial relationship with the subject company. Considering that Citi operates multiple businesses in more than 100 countries around the world, it is likely that Citi has a commercial relationship with the subject company.

Securities recommended, offered, or sold by the Firm: (i) are not insured by the Federal Deposit Insurance Corporation; (ii) are not deposits or other obligations of any insured depository institution (including Citibank); and (iii) are subject to investment risks, including the possible loss of the principal amount invested. Although information has been obtained from and is based upon sources that the Firm believes to be reliable, we do not guarantee its accuracy and it may be incomplete and condensed. Note, however, that the Firm has taken all reasonable steps to determine the accuracy and completeness of the disclosures made in the Important Disclosures section of the Product. The Firm's research department has received assistance from the subject company(ies) referred to in this Product including, but not limited to, discussions with management of the subject company(ies). Firm policy prohibits research analysts from sending draft research to subject companies. However, it should be presumed that the author of the Product has had discussions with the subject company to ensure factual accuracy prior to publication. All opinions, projections and estimates constitute the judgment of the author as of the date of the Product and these, plus any other information contained in the Product, are subject to change without notice. Prices and availability of financial instruments also are subject to change without notice. Notwithstanding other departments within the Firm advising the companies discussed in this Product, information obtained in such role is not used in the preparation of the Product. Although Citi Research does not set a predetermined frequency for publication, if the Product is a fundamental research report, it is the intention of Citi Research to provide research coverage of the/those issuer(s) mentioned therein, including in response to news affecting this issuer, subject to applicable quiet periods and capacity constraints. The Product is for informational purposes only and is not intended as an offer or solicitation for the purchase or sale of a security. Any decision to purchase securities mentioned in the Product must take into account existing public information on such security or any registered prospectus.

Investing in non-U.S. securities, including ADRs, may entail certain risks. The securities of non-U.S. issuers may not be registered with, nor be subject to the reporting requirements of the U.S. Securities and Exchange Commission. There may be limited information available on foreign securities. Foreign companies are generally not subject to uniform audit and reporting standards, practices and requirements comparable to those in the U.S. Securities of some foreign companies may be less liquid and their prices more volatile than securities of comparable U.S. companies. In addition, exchange rate movements may have an adverse effect on the value of an investment in a foreign stock and its corresponding dividend payment for U.S. investors. Net dividends to ADR investors are estimated, using withholding tax rates conventions, deemed accurate, but investors are urged to consult their tax advisor for exact dividend computations. Investors who have received the Product from the Firm may be prohibited in certain states or other jurisdictions from purchasing securities mentioned in the Product from the Firm. Please ask your Financial Consultant for additional details. Citigroup Global Markets Inc. takes responsibility for the Product in the United States. Any orders by US investors resulting from the information contained in the Product may be placed only through Citigroup Global Markets Inc.

Important Disclosures for Bell Potter Customers: Bell Potter is making this Product available to its clients pursuant to an agreement with Citigroup Global Markets Australia Pty Limited. Neither Citigroup Global Markets Australia Pty Limited nor any of its affiliates has made any determination as to the suitability of the information provided herein and clients should consult with their Bell Potter financial advisor before making any investment decision.

The Citigroup legal entity that takes responsibility for the production of the Product is the legal entity which the first named author is employed by. The Product is made available in Australia through Citigroup Global Markets Australia Pty Limited. (ABN 64 003 114 832 and AFSL No. 240992), participant of the ASX Group and regulated by the Australian Securities & Investments Commission. Citigroup Centre, 2 Park Street, Sydney, NSW 2000. The Product is made available in Australia to Private Banking wholesale clients through Citigroup Pty Limited (ABN 88 004 325 080 and AFSL 238098). Citigroup Pty Limited provides all financial product advice to Australian Private Banking wholesale clients through bankers and relationship managers. If there is any doubt about the suitability of investments held in Citigroup Private Bank accounts, investors should contact the Citigroup Private Bank in Australia. Citigroup companies may compensate affiliates and their representatives for providing products and services to clients. The Product is made available in Brazil by Citigroup Global Markets Brasil - CCTVM SA, which is regulated by CVM - Comissão de Valores Mobiliários, BACEN - Brazilian Central Bank, APIMEC - Associação dos Analistas e Profissionais de Investimento do Mercado de Capitais and ANBID - Associação Nacional dos Bancos de Investimento. Av. Paulista, 1111 - 11º andar - CEP. 01311920 - São Paulo - SP. If the Product is being made available in certain provinces of Canada by Citigroup Global Markets (Canada) Inc. ("CGM Canada"), CGM Canada has approved the Product. Citigroup Place, 123 Front Street West, Suite 1100, Toronto, Ontario M5J 2M3. This product is available in Chile through Banchile Corredores de Bolsa S.A., an indirect subsidiary of Citigroup Inc., which is regulated by the Superintendencia de Valores y Seguros. Agustinas 975, piso 2, Santiago, Chile. The Product is distributed in Germany by Citigroup Global Markets Deutschland AG ("CGMD"), which is regulated by Bundesanstalt fuer Finanzdienstleistungsaufsicht (BaFin). CGMD, Reuterweg 16, 60323 Frankfurt am Main. Research which relates to "securities" (as defined in the Securities and Futures Ordinance (Cap. 571 of the Laws of Hong Kong)) is issued in Hong Kong by, or on behalf of, Citigroup Global Markets Asia Limited which takes full responsibility for its content. Citigroup Global Markets Asia Ltd. is regulated by Hong Kong Securities and Futures Commission. If the Research is made available through Citibank, N.A., Hong Kong Branch, for its clients in Citi Private Bank, it is made available by Citibank N.A., Citibank Tower, Citibank Plaza, 3 Garden Road, Hong Kong. Citibank N.A. is regulated by the Hong Kong Monetary Authority. Please contact your Private Banker in Citibank N.A., Hong Kong, Branch if you have any queries on or any matters arising from or in connection with this document. The Product is made available in India by Citigroup Global Markets India Private Limited (CGM), which is regulated by the Securities and Exchange Board of India (SEBI), as a Research Analyst (SEBI Registration No. INH000000438). CGM is also actively involved in the business of merchant banking, stock brokerage, and depository participant, in India, and is registered with SEBI in this regard. CGM's registered office is at 1202, 12th Floor, FIFC, G Block, Bandra Kurla Complex, Bandra East, Mumbai – 400051. CGM's Corporate Identity Number is U99999MH2000PTC126657, and its contact details are: Tel:+9102261759999 Fax:+9102261759961. The Product is made available in Indonesia through PT Citigroup Securities Indonesia. 5/F, Citibank Tower, Bapindo Plaza, Jl. Jend. Sudirman Kav. 54-55, Jakarta 12190. Neither this Product nor any copy hereof may be distributed in Indonesia or to any Indonesian citizens wherever they are domiciled or to Indonesian residents except in compliance with applicable capital market laws and regulations. This Product is not an offer of securities in Indonesia. The securities referred to in this Product have not been registered with the Capital Market and Financial Institutions Supervisory Agency (BAPEPAM-LK) pursuant to relevant capital market laws and regulations, and may not be offered or sold within the territory of the Republic of Indonesia or to Indonesian citizens through a public offering or in circumstances which constitute an offer within the meaning of the Indonesian capital market laws and regulations. The Product is made available in Israel through Citibank NA, regulated by the Bank of Israel and the Israeli Securities Authority. Citibank, N.A, Platinum Building, 21 Ha'arba'ah St, Tel Aviv, Israel. The Product is made available in Italy by Citigroup Global Markets Limited, which is authorised by the PRA and regulated by the FCA and the PRA. Via dei Mercanti, 12, Milan, 20121, Italy. The Product is made available in Japan by Citigroup Global Markets Japan Inc. ("CGMJ"), which is regulated by Financial Services Agency, Securities and Exchange Surveillance Commission, Japan Securities Dealers Association, Tokyo Stock Exchange and Osaka Securities Exchange. Shin-Marunouchi Building, 1-5-1 Marunouchi, Chiyoda-ku, Tokyo 100-6520 Japan. If the Product was distributed by SMBC Nikko Securities Inc. it is being so distributed under license. In the event that an error is found in an CGMJ research report, a revised version will be posted on the Firm's Citi Velocity website. If you have questions regarding Citi Velocity, please call (81 3) 6270-3019 for help. The Product is made available in Korea by Citigroup Global Markets Korea Securities Ltd., which is regulated by the Financial Services Commission, the Financial Supervisory Service and the Korea Financial Investment Association (KOFIA). Citibank Building, 39 Da-dong, Jung-gu, Seoul 100-180, Korea. KOFIA makes available registration information of research analysts on its website. Please visit the following website if you wish to find KOFIA registration information on research analysts of Citigroup Global Markets Korea Securities

Ltd. http://dis.kofia.or.kr/websquare/index.jsp?w2xPath=/wq/fundMgr/DISFundMgrAnalystList.xml&divisionId=MDIS03002002000000&serviceId=SDIS03002002000. The Product is made available in Korea by Citibank Korea Inc., which is regulated by the Financial Services Commission and the Financial Supervisory Service. Address is Citibank Building, 39 Da-dong, Jung-gu, Seoul 100-180, Korea. The Product is made available in Malaysia by Citigroup Global Markets Malaysia Sdn Bhd (Company No. 460819-D) ("CGMM") to its clients and CGMM takes responsibility for its contents. CGMM is regulated by the Securities Commission of Malaysia. Please contact CGMM at Level 43 Menara Citibank, 165 Jalan Ampang, 50450 Kuala Lumpur, Malaysia in respect of any matters arising from, or in connection with, the Product. The Product is made available in Mexico by Acciones y Valores Banamex, S.A. De C. V., Casa de Bolsa, Integrante del Grupo Financiero Banamex ("Accival") which is a wholly owned subsidiary of Citigroup Inc. and is regulated by Comision Nacional Bancaria y de Valores. Reforma 398, Col. Juarez, 06600 Mexico, D.F. In New Zealand the Product is made available to 'wholesale clients' only as defined by s5C(1) of the Financial Advisers Act 2008 ('FAA') through Citigroup Global Markets Australia Pty Ltd (ABN 64 003 114 832 and AFSL No. 240992), an overseas financial adviser as defined by the FAA, participant of the ASX Group and regulated by the Australian Securities & Investments Commission. Citigroup Centre, 2 Park Street, Sydney, NSW 2000. The Product is made available in Pakistan by Citibank N.A. Pakistan branch, which is regulated by the State Bank of Pakistan and Securities Exchange Commission, Pakistan. AWT Plaza, 1.1. Chundrigar Road, P.O. Box 4889, Karachi-74200. The Product is made available in the Philippines through Citicorp Financial Services and Insurance Brokerage Philippines, Inc., which is regulated by the Philippines Securities and Exchange Commission. 20th Floor Citibank Square Bldg. The Product is made available in the Philippines through Citibank NA Philippines branch, Citibank Tower, 8741 Paseo De Roxas, Makati City, Manila. Citibank NA Philippines NA is regulated by The Bangko Sentral ng Pilipinas. The Product is made available in Poland by Dom Maklerski Banku Handlowego SA an indirect subsidiary of Citigroup Inc., which is regulated by Komisja Nadzoru Finansowego. Dom Maklerski Banku Handlowego S.A. ul.Senatorska 16, 00-923 Warszawa. The Product is made available in the Russian Federation through ZAO Citibank, which is licensed to carry out banking activities in the Russian Federation in accordance with the general banking license issued by the Central Bank of the Russian Federation and brokerage activities in accordance with the license issued by the Federal Service for Financial Markets. Neither the Product nor any information contained in the Product shall be considered as advertising the securities mentioned in this report within the territory of the Russian Federation or outside the Russian Federation. The Product does not constitute an appraisal within the meaning of the Federal Law of the Russian Federation of 29 July 1998 No. 135-FZ (as amended) On Appraisal Activities in the Russian Federation. 8-10 Gasheka Street, 125047 Moscow. The Product is made available in Singapore through Citigroup Global Markets Singapore Pte. Ltd. ("CGMSPL"), a capital markets services license holder, and regulated by Monetary Authority of Singapore. Please contact CGMSPL at 8 Marina View, 21st Floor Asia Square Tower 1, Singapore 018960, in respect of any matters arising from, or in connection with, the analysis of this document. This report is intended for recipients who are accredited, expert and institutional investors as defined under the Securities and Futures Act (Cap. 289). The Product is made available by The Citigroup Private Bank in Singapore through Citibank, N.A., Singapore Branch, a licensed bank in Singapore that is regulated by Monetary Authority of Singapore. Please contact your Private Banker in Citibank N.A., Singapore Branch if you have any queries on or any matters arising from or in connection with this document. This report is intended for recipients who are accredited, expert and institutional investors as defined under the Securities and Futures Act (Cap. 289). This report is distributed in Singapore by Citibank Singapore Ltd ("CSL") to selected Citigold/Citigold Private Clients. CSL provides no independent research or analysis of the substance or in preparation of this report. Please contact your Citigold//Citigold Private Client Relationship Manager in CSL if you have any queries on or any matters arising from or in connection with this report. This report is intended for recipients who are accredited investors as defined under the Securities and Futures Act (Cap. 289). Citigroup Global Markets (Pty) Ltd. is incorporated in the Republic of South Africa (company registration number 2000/025866/07) and its registered office is at 145 West Street, Sandton, 2196, Saxonwold. Citigroup Global Markets (Pty) Ltd. is regulated by JSE Securities Exchange South Africa, South African Reserve Bank and the Financial Services Board. The investments and services contained herein are not available to private customers in South Africa. The Product is made available in the Republic of China through Citigroup Global Markets Taiwan Securities Company Ltd. ("CGMTS"), 14 and 15F, No. 1, Songzhi Road, Taipei 110, Taiwan and/or through Citibank Securities (Taiwan) Company Limited ("CSTL"), 14 and 15F, No. 1, Songzhi Road, Taipei 110, Taiwan, subject to the respective license scope of each entity and the applicable laws and regulations in the Republic of China. CGMTS and CSTL are both regulated by the Securities and Futures Bureau of the Financial Supervisory Commission of Taiwan, the Republic of China. No portion of the Product may be reproduced or quoted in the Republic of China by the press or any third parties [without the written authorization of CGMTS and CSTL]. If the Product covers securities which are not allowed to be offered or traded in the Republic of China, neither the Product nor any information contained in the Product shall be considered as advertising the securities or making recommendation of the securities in the Republic of China. The Product is for informational purposes only and is not intended as an offer or solicitation for the purchase or sale of a security or financial products. Any decision to purchase securities or financial products mentioned in the Product must take into account existing public information on such security or the financial products or any registered prospectus. The Product is made available in Thailand through Citicorp Securities (Thailand) Ltd., which is regulated by the Securities and Exchange Commission of Thailand. 399 Interchange 21 Building, 18th Floor, Sukhumvit Road, Klongtoey Nua, Wattana ,Bangkok 10110, Thailand. The Product is made available in Turkey through Citibank AS which is regulated by Capital Markets Board. Tekfen Tower, Eski Buyukdere Caddesi # 209 Kat 2B, 23294 Levent, Istanbul, Turkey. In the U.A.E, these materials (the "Materials") are communicated by Citigroup Global Markets Limited, DIFC branch ("CGML"), an entity registered in the Dubai International Financial Center ("DIFC") and licensed and regulated by the Dubai Financial Services Authority ("DFSA") to Professional Clients and Market Counterparties only and should not be relied upon or distributed to Retail Clients. A distribution of the different Citi Research ratings distribution, in percentage terms for Investments in each sector covered is made available on request. Financial products and/or services to which the Materials relate will only be made available to Professional Clients and Market Counterparties. The Product is made available in United Kingdom by Citigroup Global Markets Limited, which is authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA. This material may relate to investments or services of a person outside of the UK or to other matters which are not authorised by the PRA nor regulated by the FCA and the PRA and further details as to where this may be the case are available upon request in respect of this material. Citigroup Centre, Canada Square, Canary Wharf, London, E14 5LB. The Product is made available in United States by Citigroup Global Markets Inc, which is a member of FINRA and registered with the US Securities and Exchange Commission. 388 Greenwich Street, New York, NY 10013. Unless specified to the contrary, within EU Member States, the Product is made available by Citigroup Global Markets Limited, which is authorised by the PRA and regulated by the FCA and the PRA.

The Product is not to be construed as providing investment services in any jurisdiction where the provision of such services would not be permitted.

Subject to the nature and contents of the Product, the investments described therein are subject to fluctuations in price and/or value and investors may get back less than originally invested. Certain high-volatility investments can be subject to sudden and large falls in value that could equal or exceed the amount invested. Certain investments contained in the Product may have tax implications for private customers whereby levels and basis of taxation may be subject to change. If in doubt, investors should seek advice from a tax adviser. The Product does not purport to identify the nature of the specific market or other risks associated with a particular transaction. Advice in the Product is general and should not be construed as personal advice given it has been prepared without taking account of the objectives, financial situation or needs of any particular investor. Accordingly, investors should, before acting on the advice, consider the appropriateness of the advice, having regard to their objectives, financial situation and needs. Prior to acquiring any financial product, it is the client's responsibility to obtain the relevant offer document for the product and consider it before making a decision as to whether to purchase the product.

Citi Research product may source data from dataCentral. dataCentral is a Citi Research proprietary database, which includes Citi estimates, data from company reports and feeds from Thomson Reuters. The printed and printable version of the research report may not include all the information (e.g., certain financial summary information and comparable company data) that is linked to the online version available on Citi's proprietary electronic distribution platforms.

the linked site does not in anyway form part of this document. Accessing such website or following such link through the Product or the website of the Firm shall be at your own risk and the Firm shall have no liability arising out of, or in connection with, any such referenced website.

ADDITIONAL INFORMATION IS AVAILABLE UPON REQUEST